2-Colouring Shift-chains

Zak Smith

Andrzej Antoni Kwaśniewski

Tutor: dr hab. Piotr Micek

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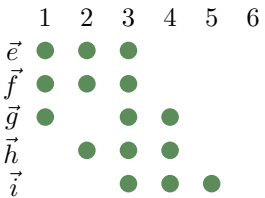


Shift-chains

Given an r-uniform hypergraph $H = (V, E)$ and a total ordering $<_V$ on V, each edge e can be interpreted as a vector: $\vec{e} = \{ e_1, \ldots, e_r \}$

where $e_1 < e_2 < \cdots < e_r$.

A gap of $\vec{e}$ is $(e_i, e_{i+1}), 1 \leq i < r$.



We define a partial order on edges by

$$\vec{e} <_E \vec{f} \iff \forall_i \, e_i \leq f_i.$$

$\vec{e}$ and $\vec{f}$ are comparable if either

$$\vec{e} <_E \vec{f} \text{ or } \vec{f} <_E \vec{e}$$

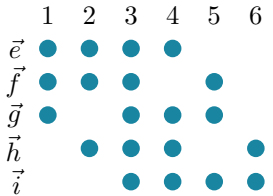
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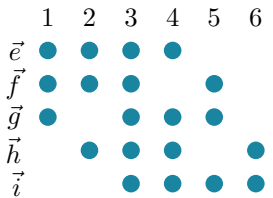
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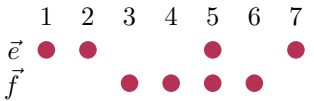
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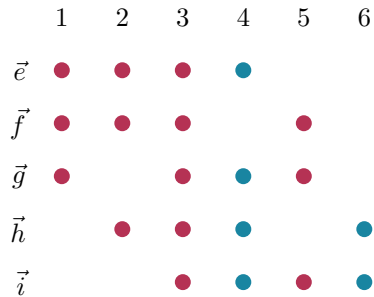




Not a shift-chain as $e_1 < f_1 \wedge e_4 > f_4$

Colouring a shift-chain

A **2-colouring** gives each vertex one of two colours.

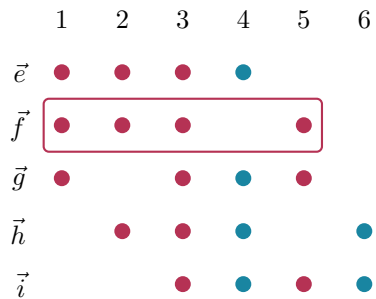


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A **2-colouring** gives each vertex one of two colours.

It is **proper** if every edge contains both colours.

$\vec{f}$ is all red: this colouring is not proper.

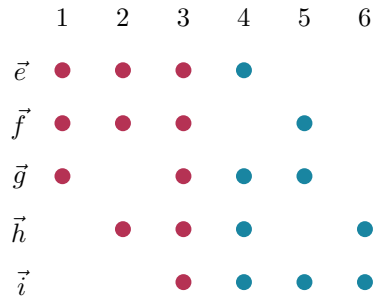


Colouring a shift-chain

A **2-colouring** gives each vertex one of two colours.

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Colour vertex 5 blue: now every edge is mixed.



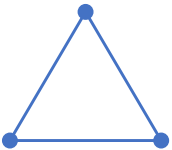
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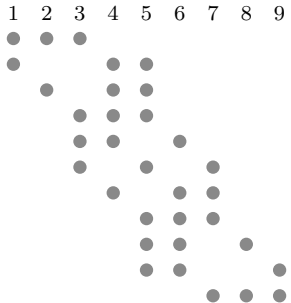
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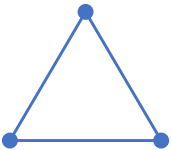
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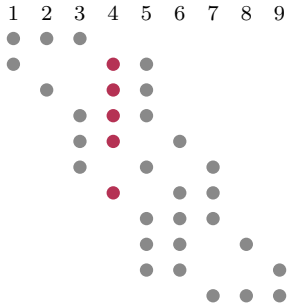
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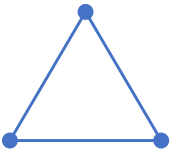
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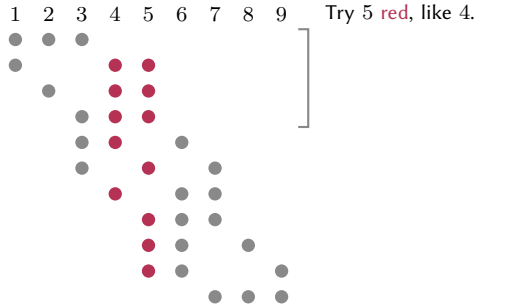
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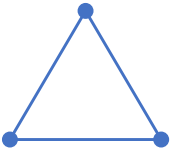
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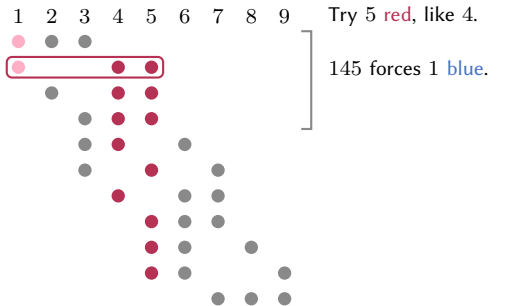
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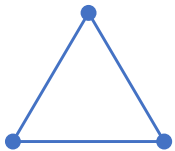
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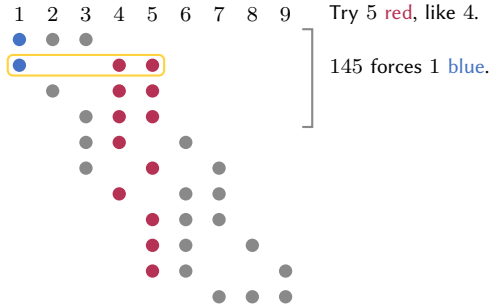
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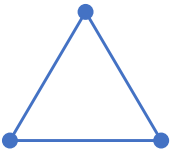
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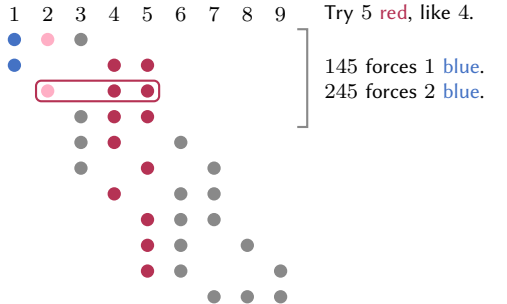
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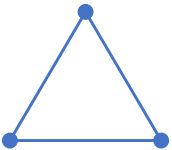
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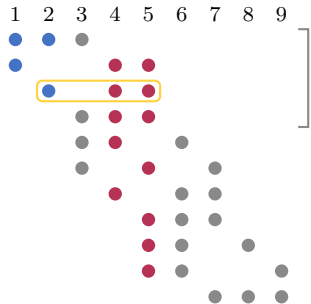
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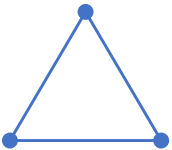
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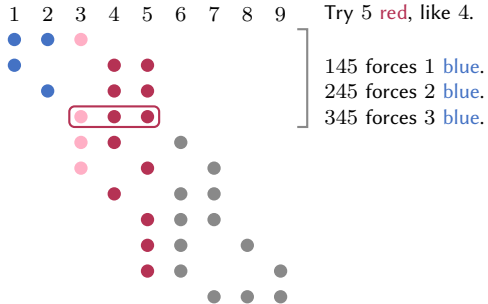
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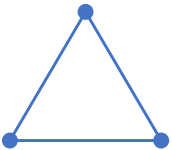
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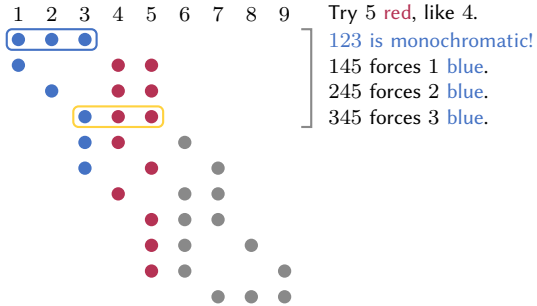
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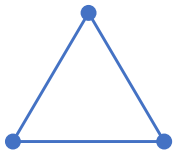
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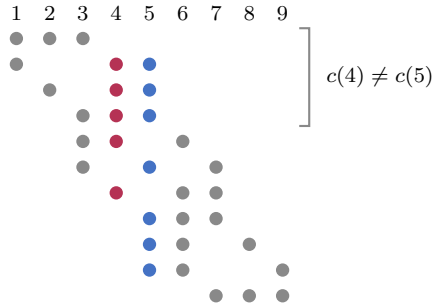
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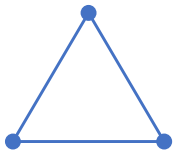
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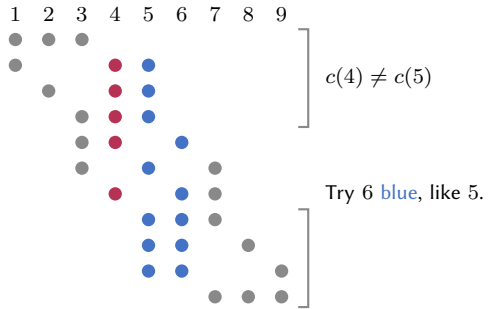
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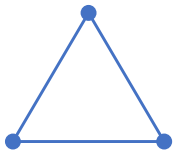
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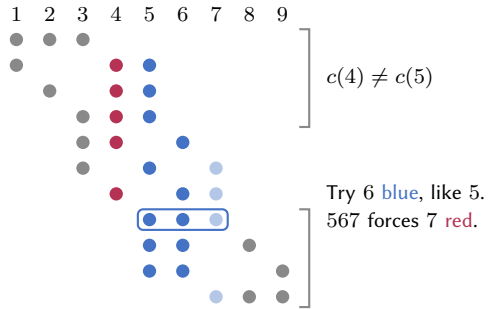
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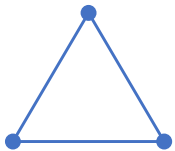
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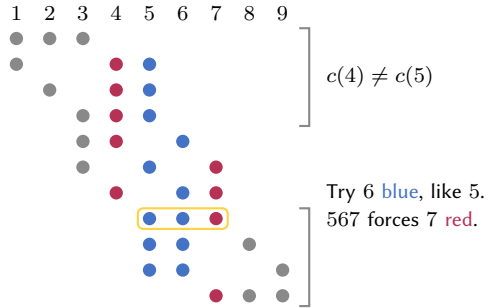
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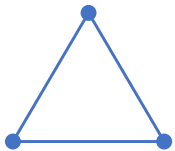
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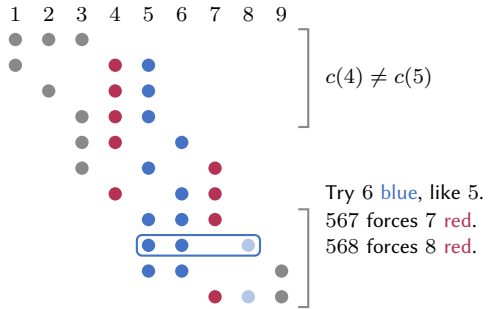
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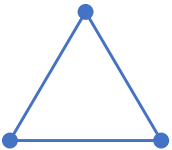
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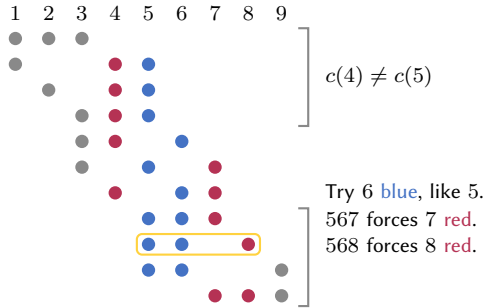
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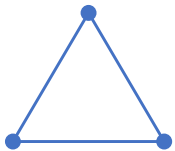
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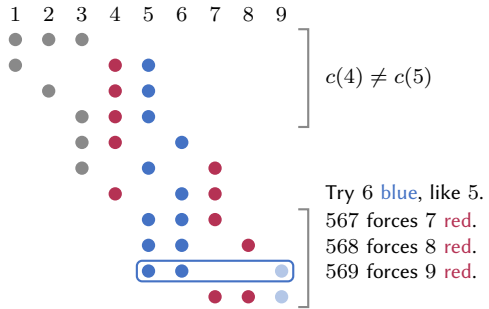
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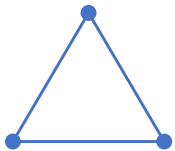
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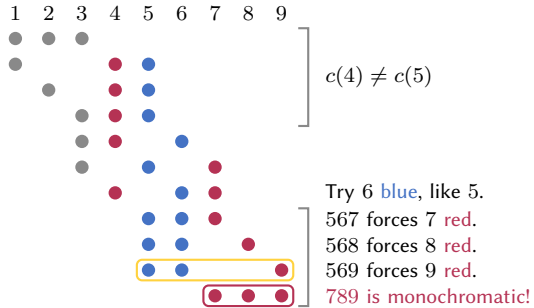
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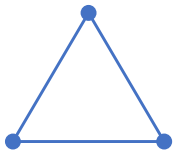
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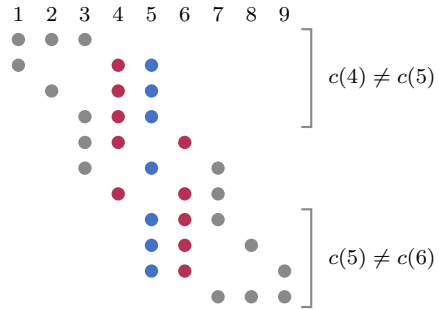
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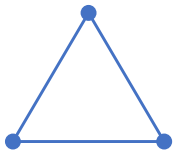
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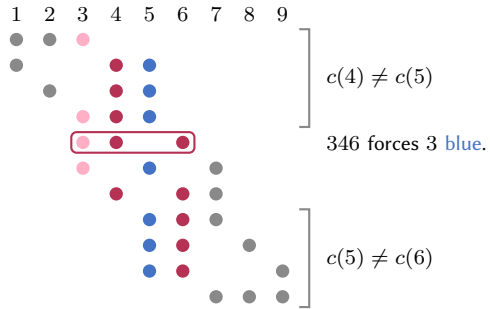
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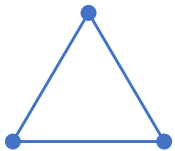
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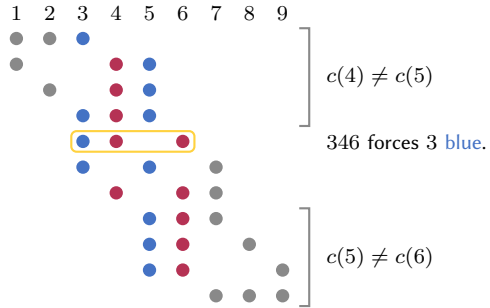
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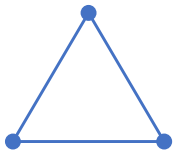
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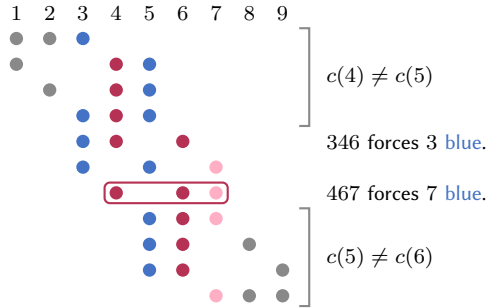
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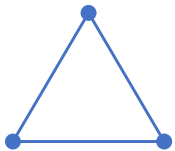
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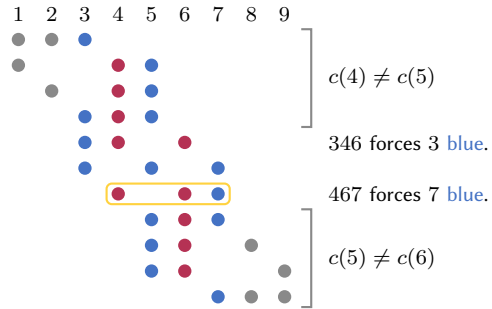
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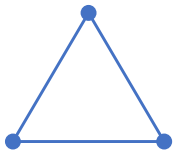
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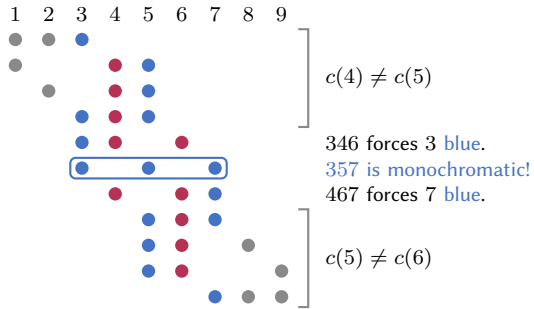
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$\{3, 5, 7\}$ is forced monochromatic: no proper 2-colouring.

From Pálvölgyi's question to Smith's theorem

Pálvölgyi's question (2010)

Are all r-shift-chains 2-colourable when r is large enough?

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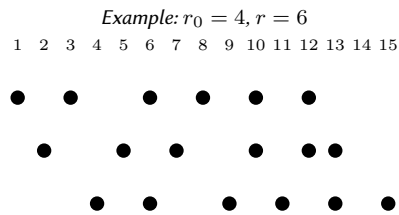
Theorem (Smith, 2026)
All r-shift-chains of uniformity at least $r_0 = 4$ are properly 2-colourable.

Claim

If every r_0-shift-chain is properly k-colourable, then every r-shift-chain with $r > r_0$ is properly k-colourable.

Proof

Example: $r_0 = 4, r = 6$



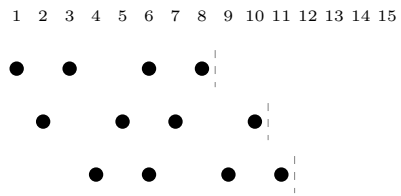
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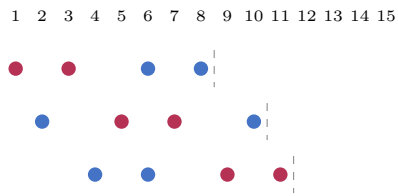
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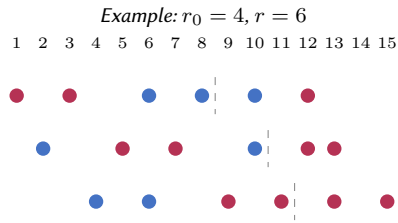


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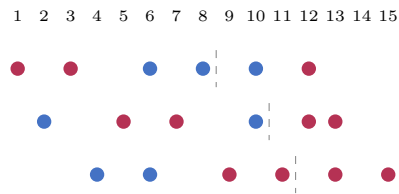
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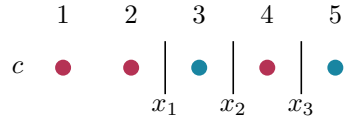
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By the claim, it suffices to prove this for 4-shift-chains.

2-colourings and switches

A **2-colouring** assigns each vertex red or blue.
A **switch** lies between consecutive vertices of different colours. Fixing the first colour, the switches determine c.



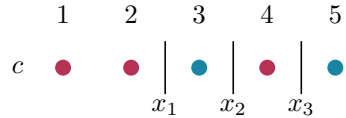
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$$u < x \iff u \le i, \qquad u > x \iff u \ge i+1.$$



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$$2 < x_1 < 3 < x_2 < 4 < x_3 < 5.$$

$$3 < x_2, \quad 4 > x_2.$$

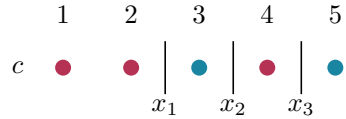
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v^- and v^+ denote the switch positions immediately before and after vertex v.



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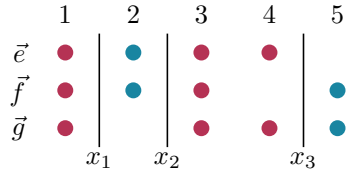
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Switch sets and well-spread colourings

Edge $\vec{e}$ **contains** switch x if $e_i < x < e_{i+1}$ for some i.



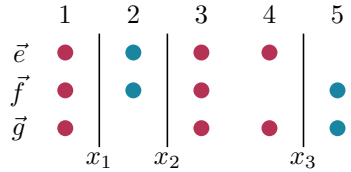
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The **switch set** $S_c(\vec{e})$ consists of all switches contained in $\vec{e}$.

Its size $|S_c(\vec{e})|$ is the **switch count**.



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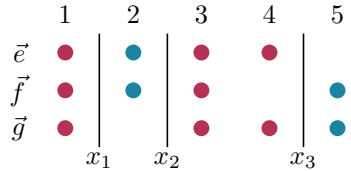
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Colouring c is k-**well-spread** if every edge satisfies

$$1 \leq |S_c(\vec{e})| \leq k.$$



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Switch counts: $2, 3, 3$.

This colouring is **3-well-spread**.

Monochromatic edges: even switches in every gap

Each switch flips the colour: ● $\xrightarrow{x_1}$ ● $\xrightarrow{x_2}$ ●. Two flips restore the colour.



Gaps: $(2, 0, 0)$ — monochromatic.

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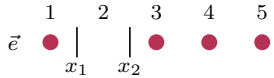
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An $(i, i+1)$-**edge** has exactly two switches, both in the i-th gap.

The three types of $(i, i+1)$-edges

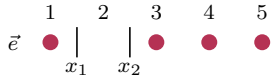
An edge is monochromatic iff every gap has an even switch count.
A colouring is 2-well-spread iff every edge has one or two switches.



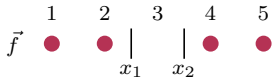
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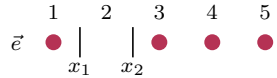
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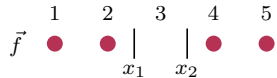
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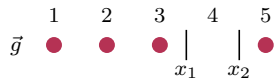
In a 2-well-spread colouring, an edge is monochromatic iff it is a $(1,2)$-, $(2,3)$-, or $(3,4)$-edge.



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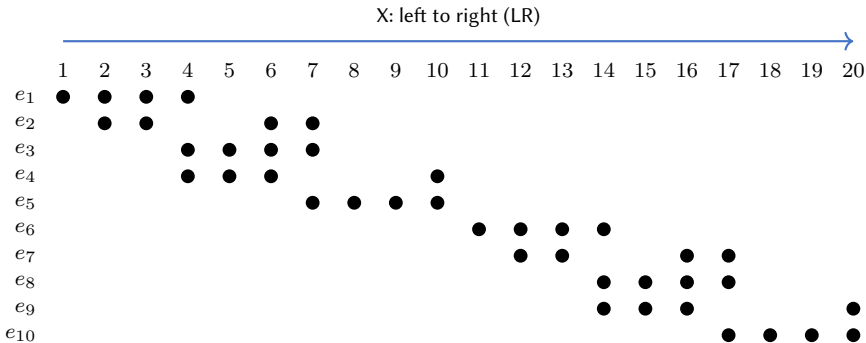
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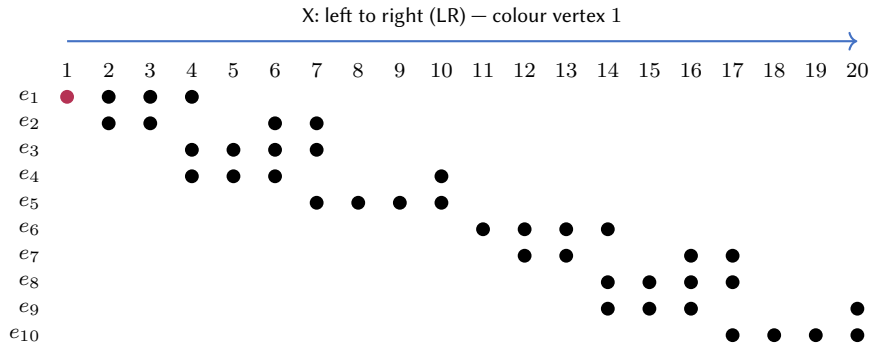
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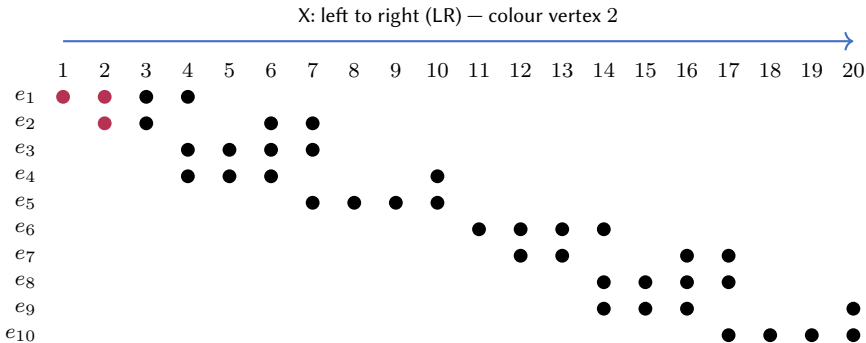
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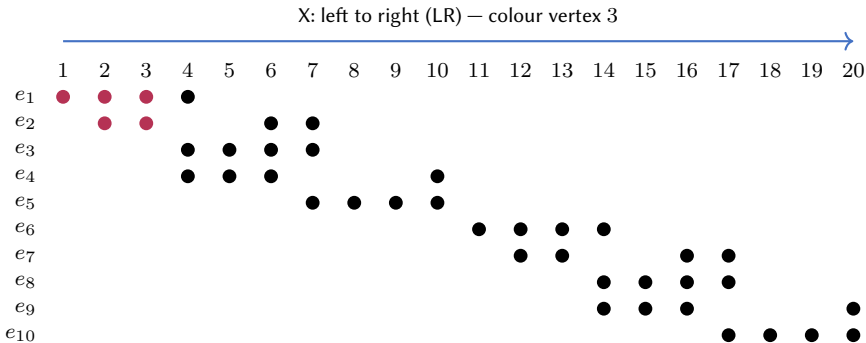
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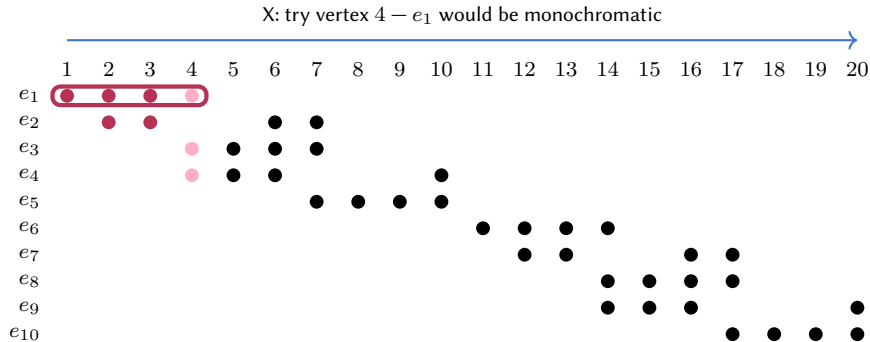
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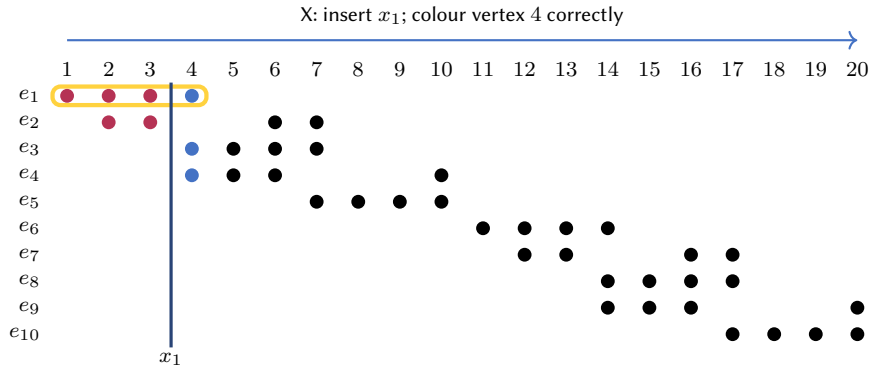
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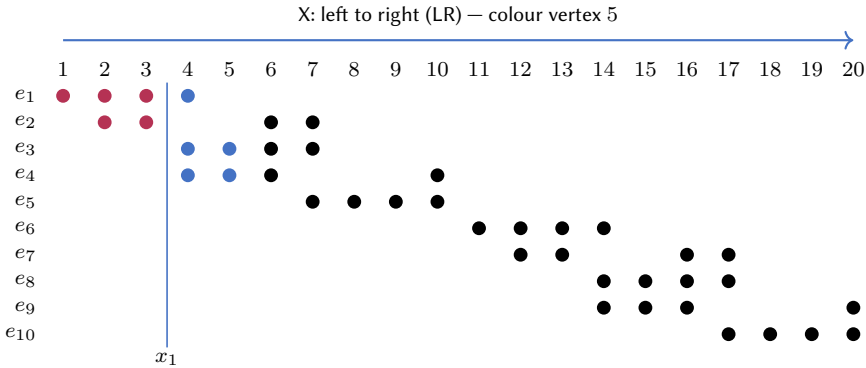
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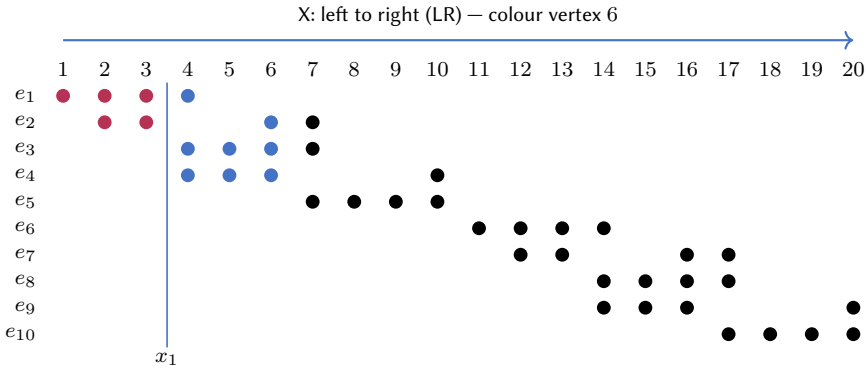
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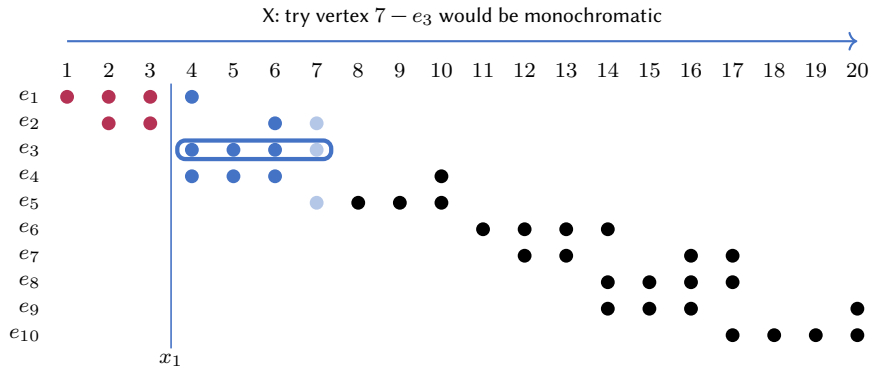
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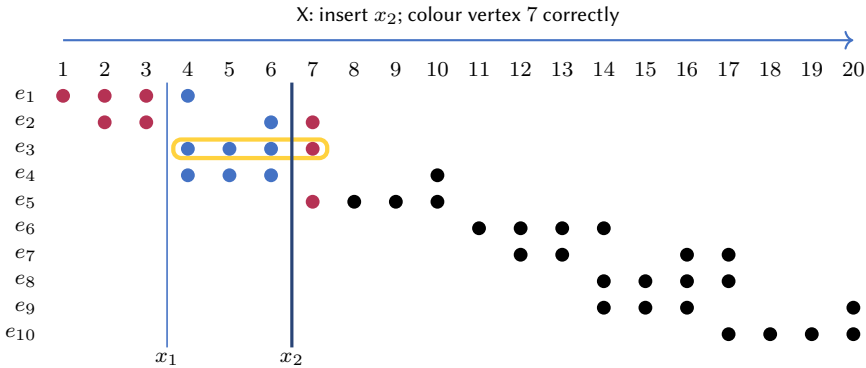
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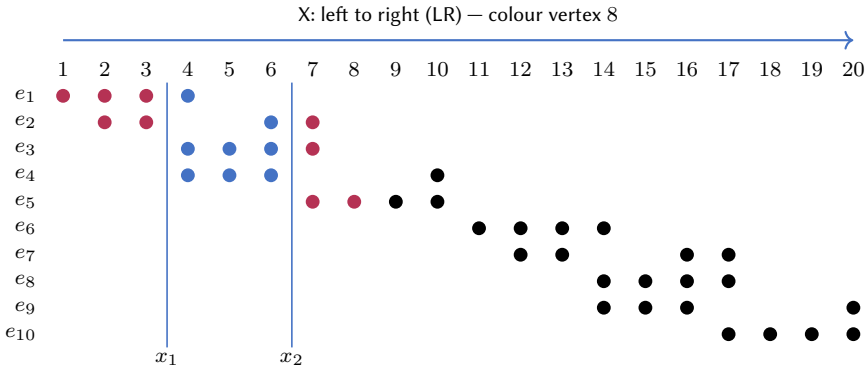
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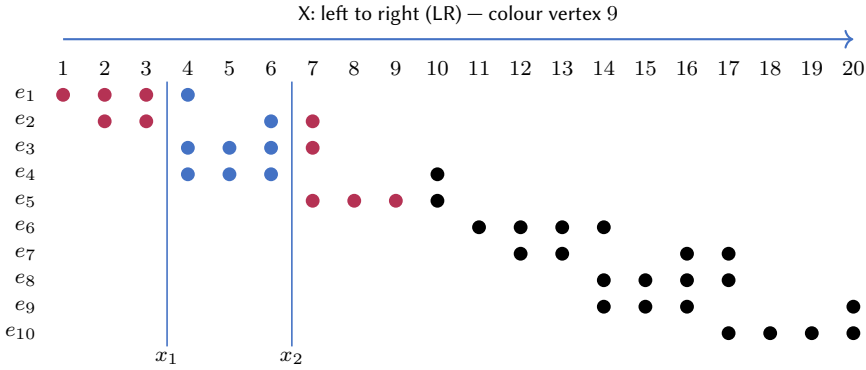
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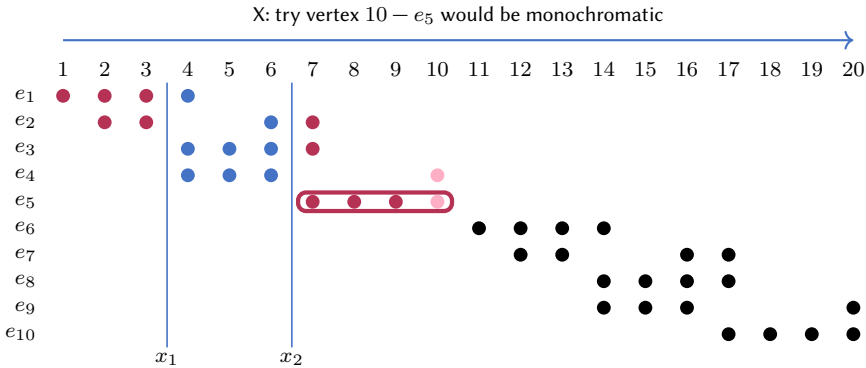
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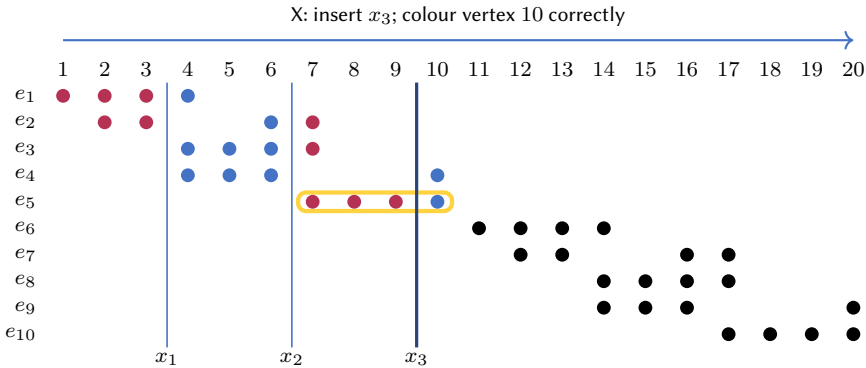
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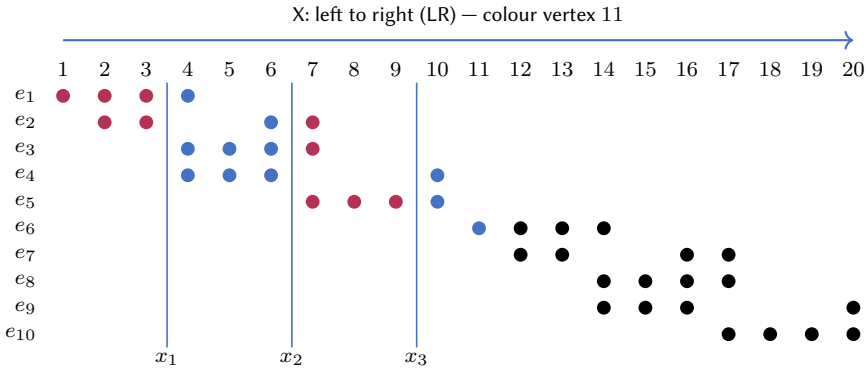
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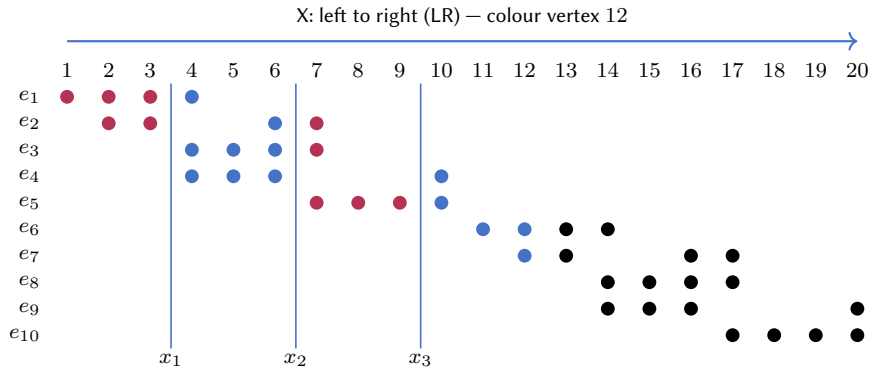
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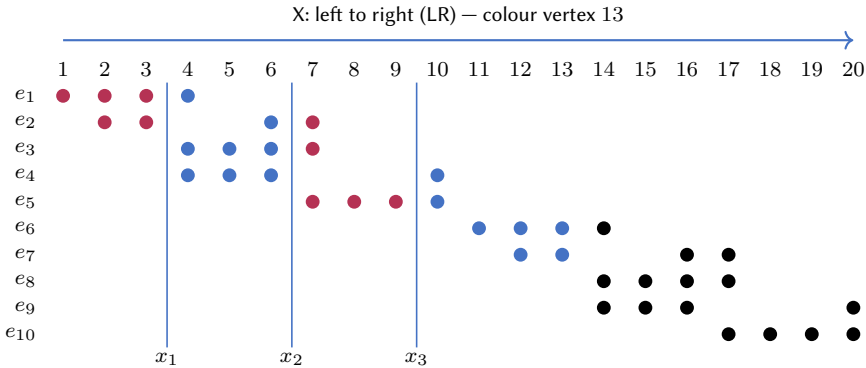
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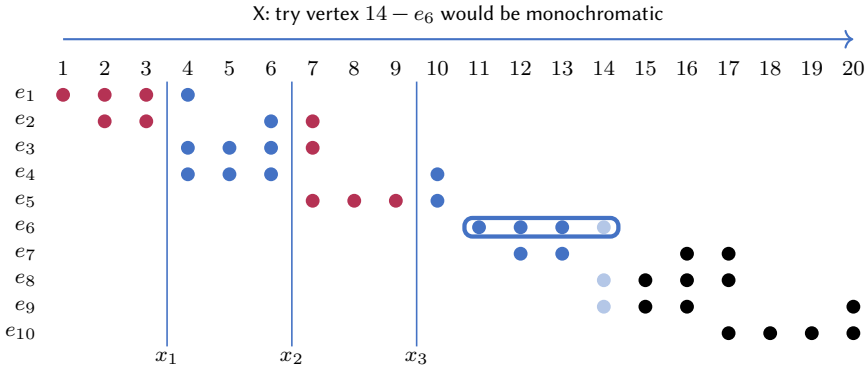
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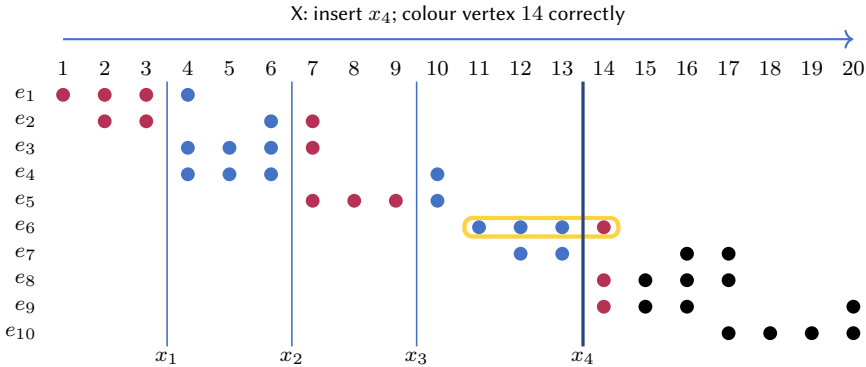
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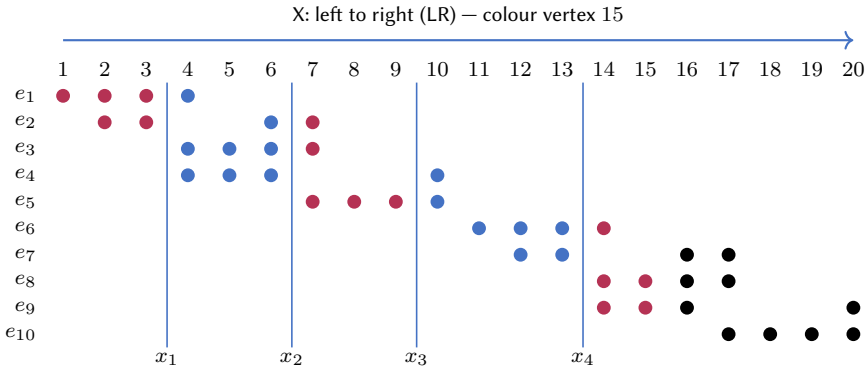
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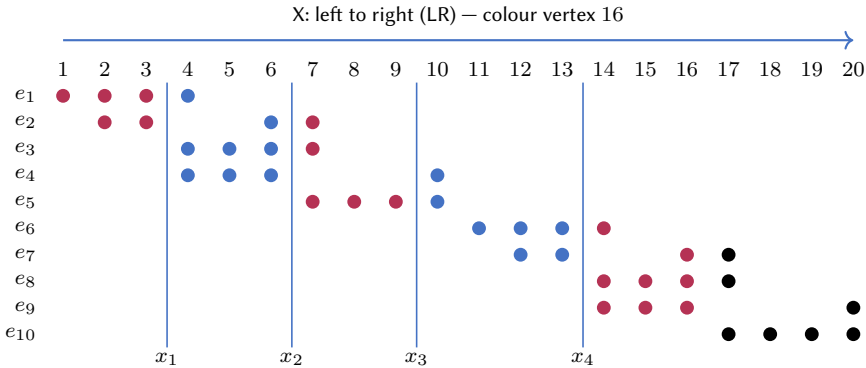
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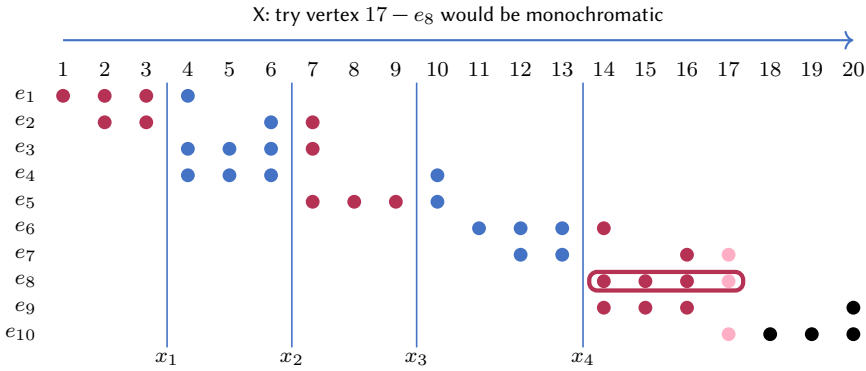
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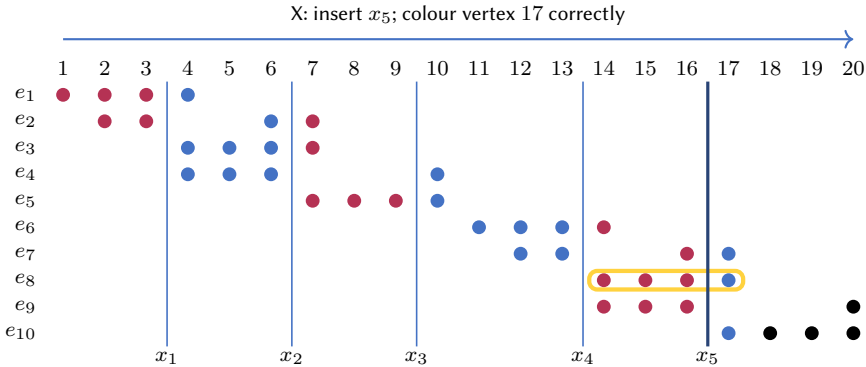
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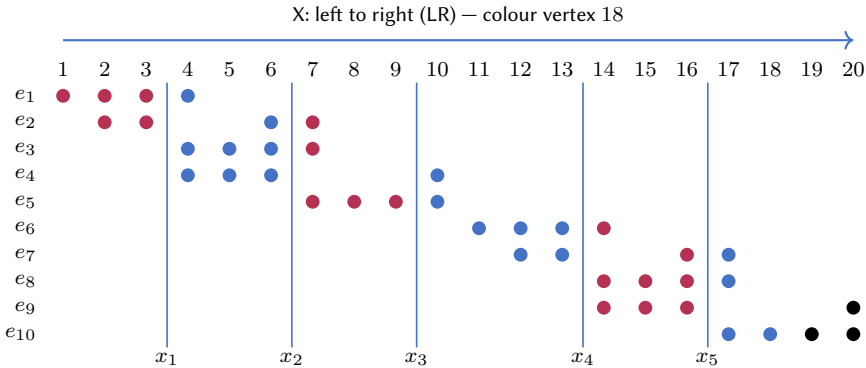
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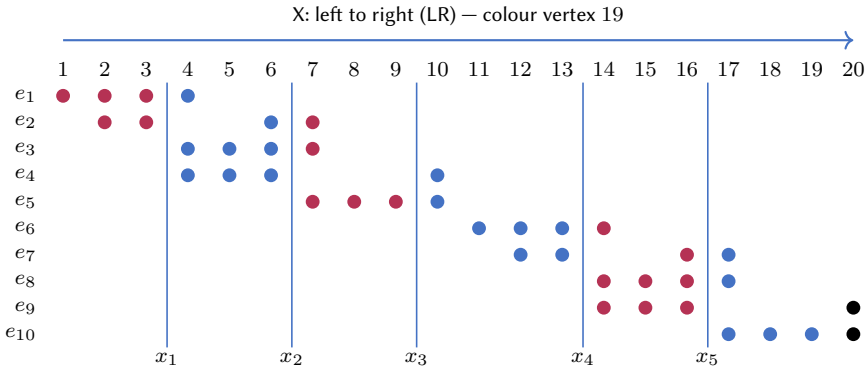
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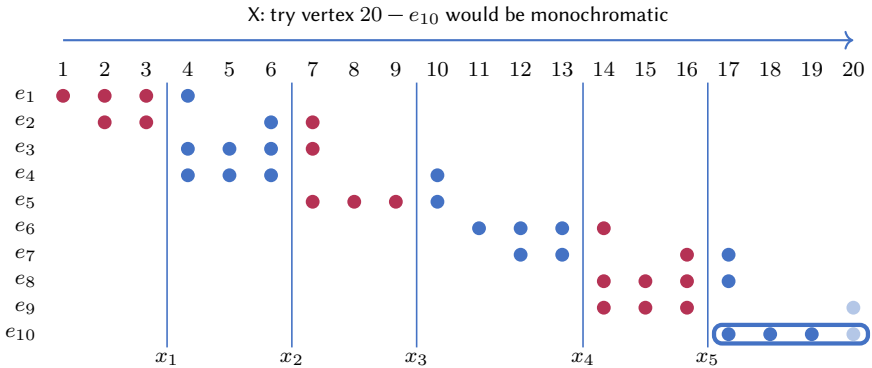
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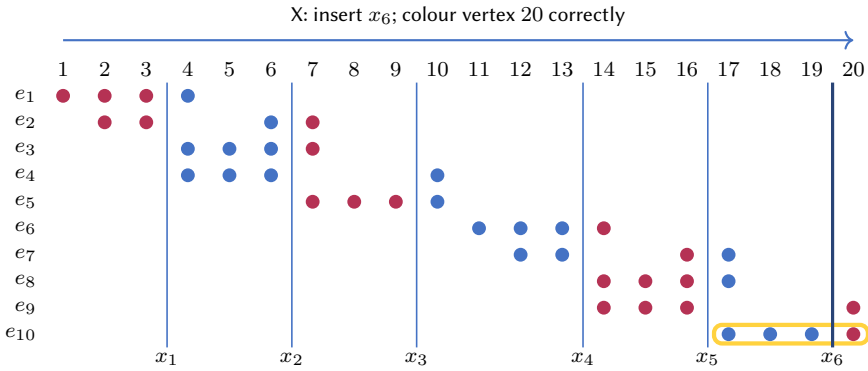
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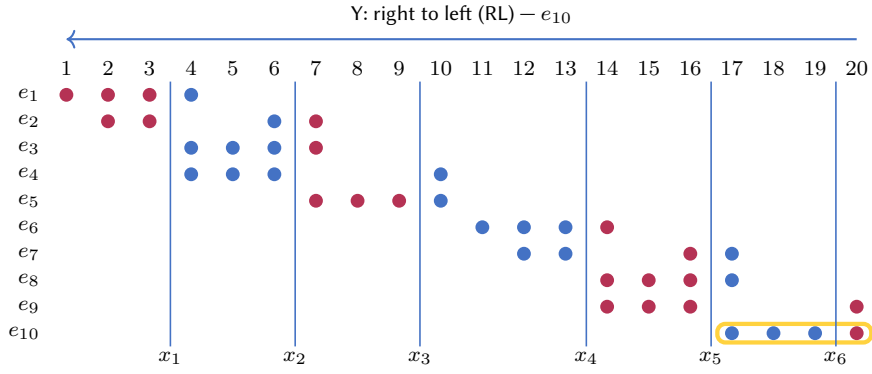
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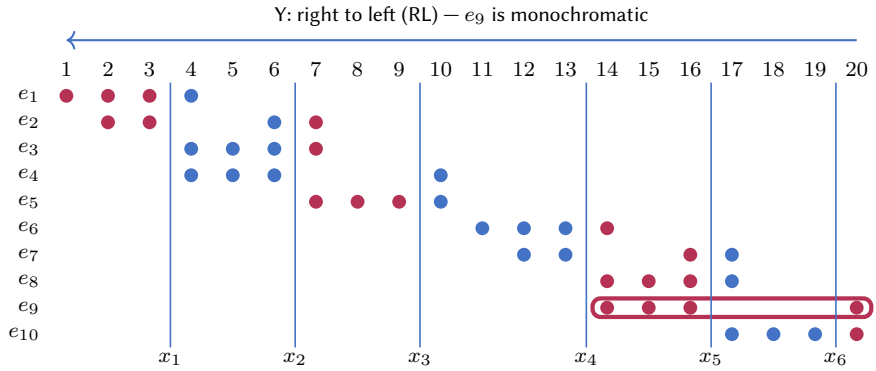
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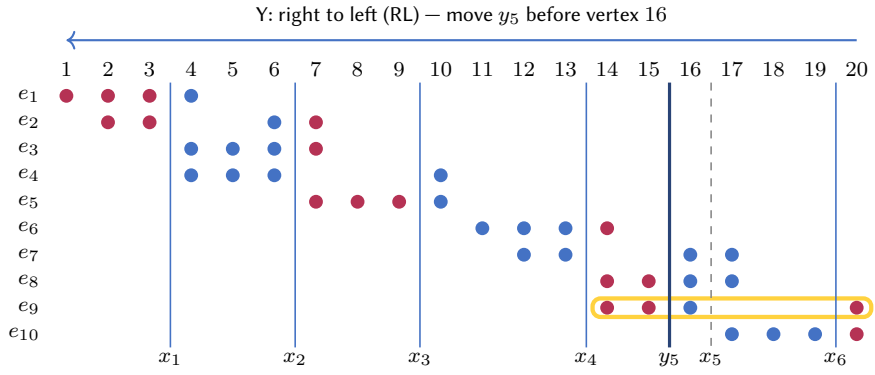
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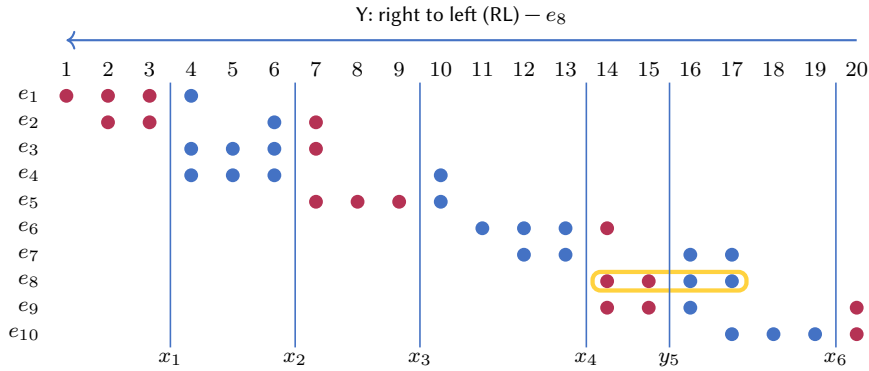
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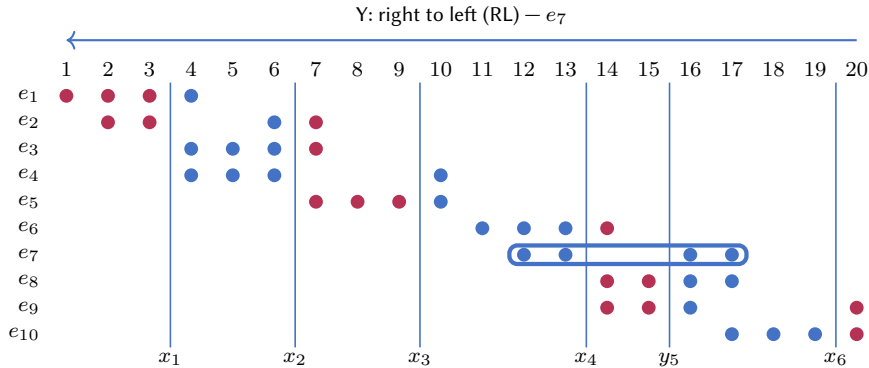
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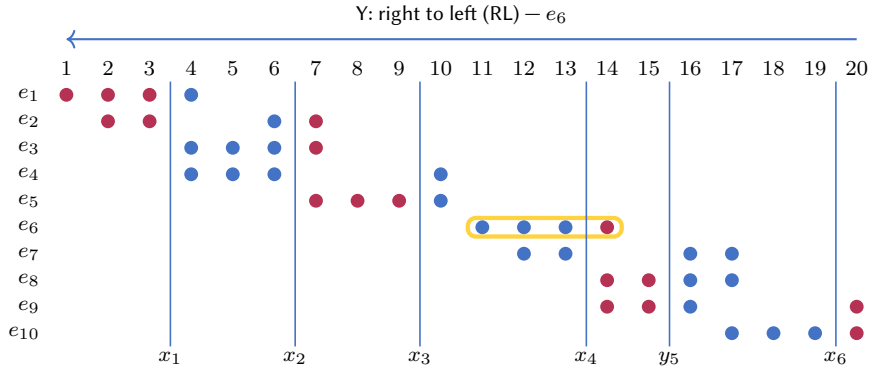
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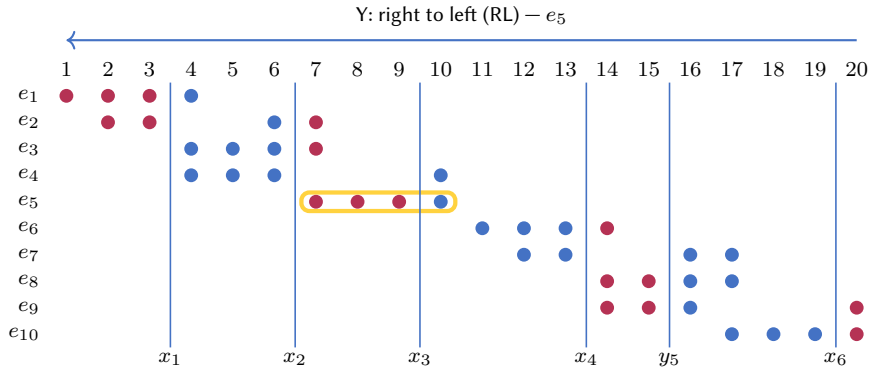
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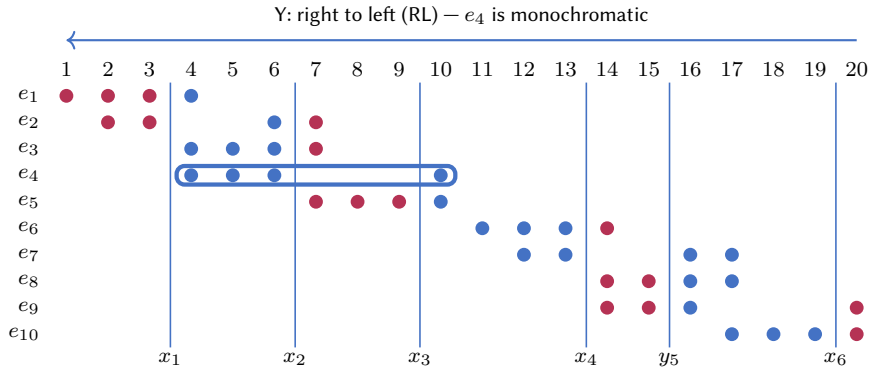
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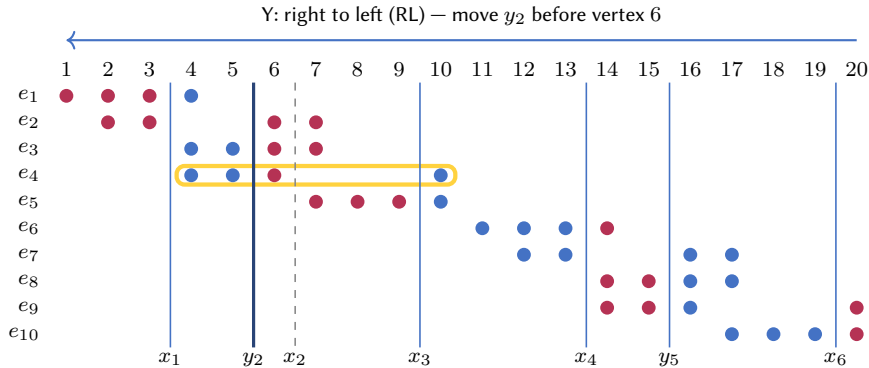
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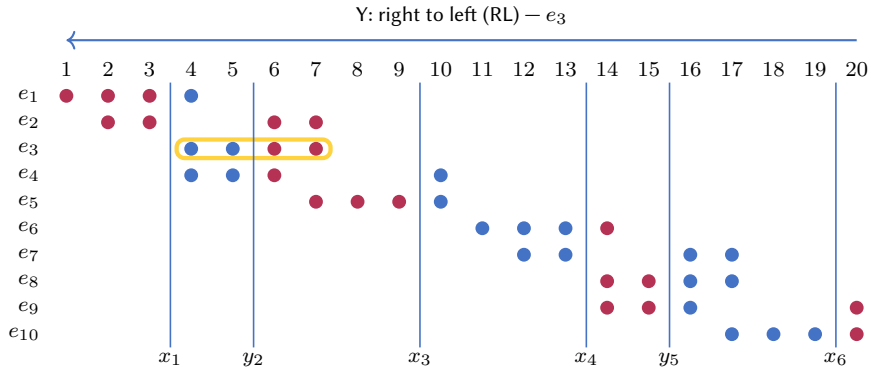
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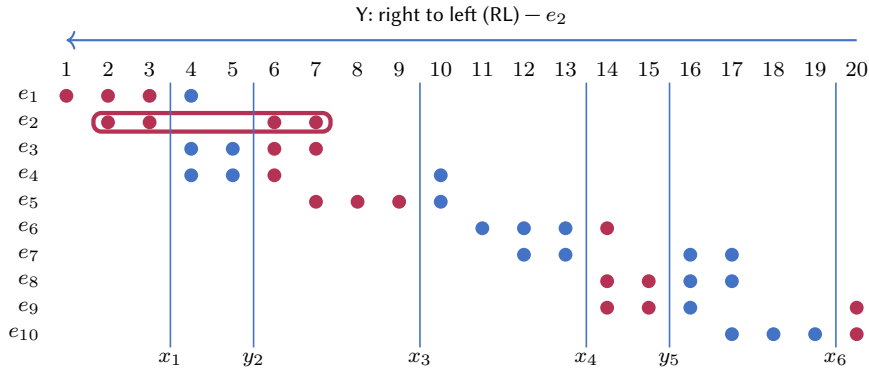
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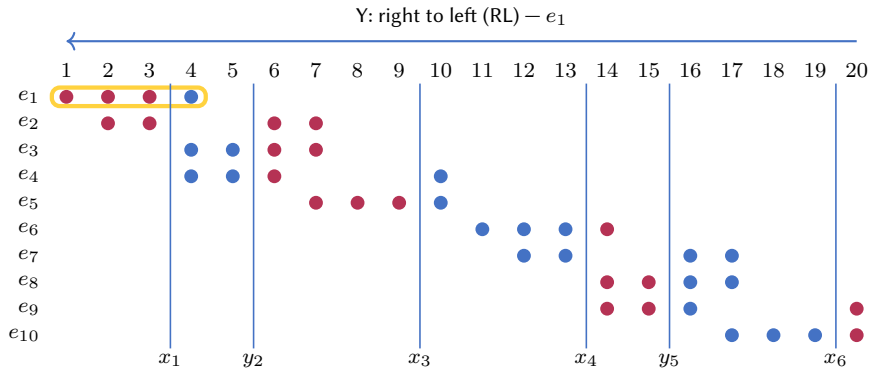
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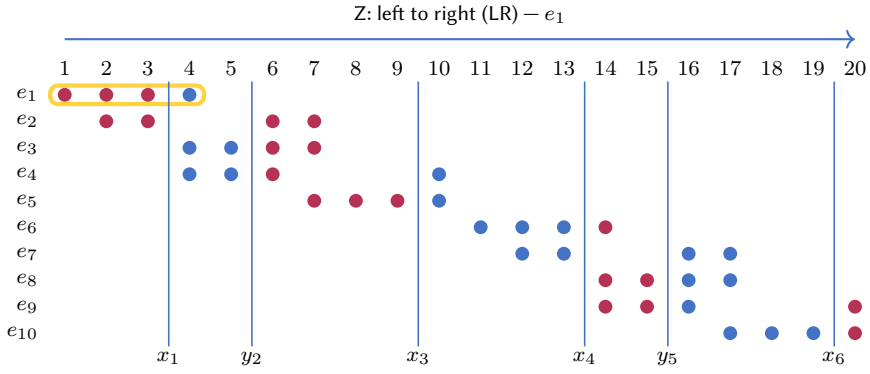


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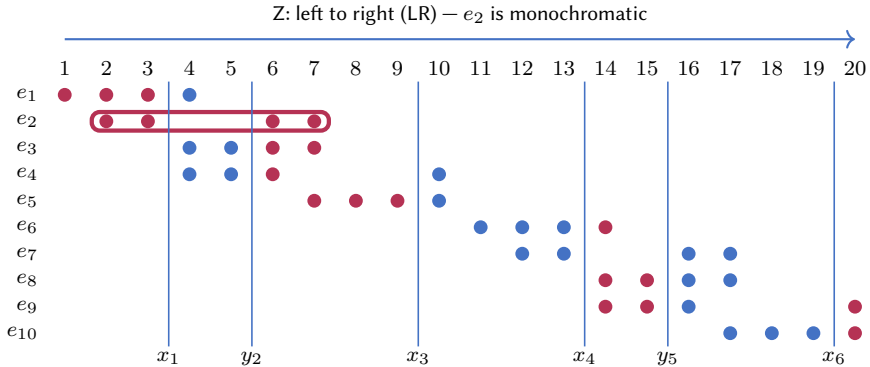


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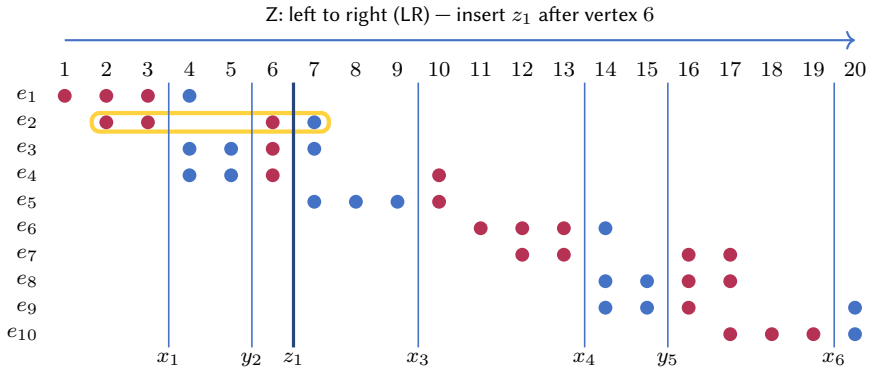


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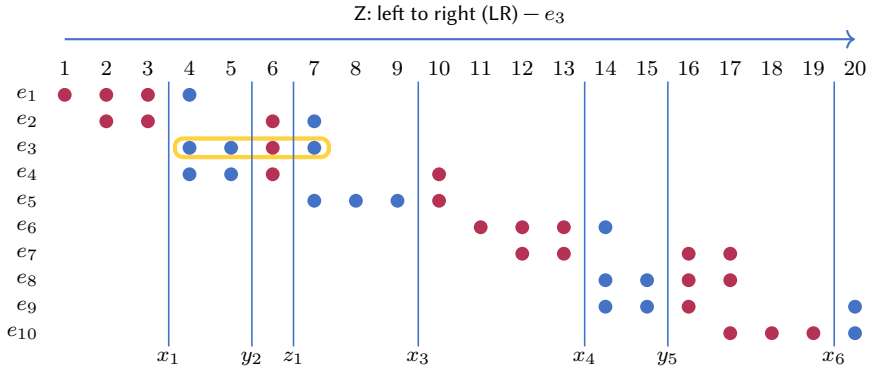


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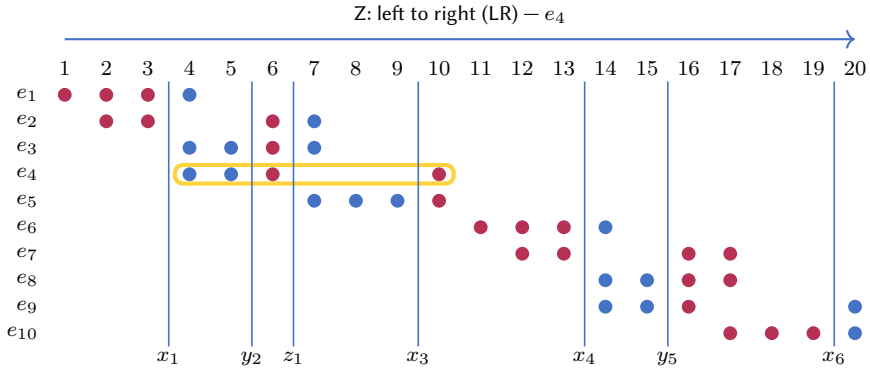


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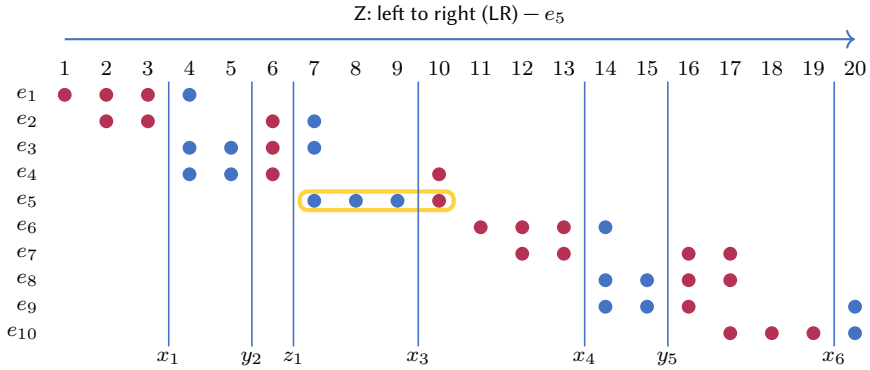


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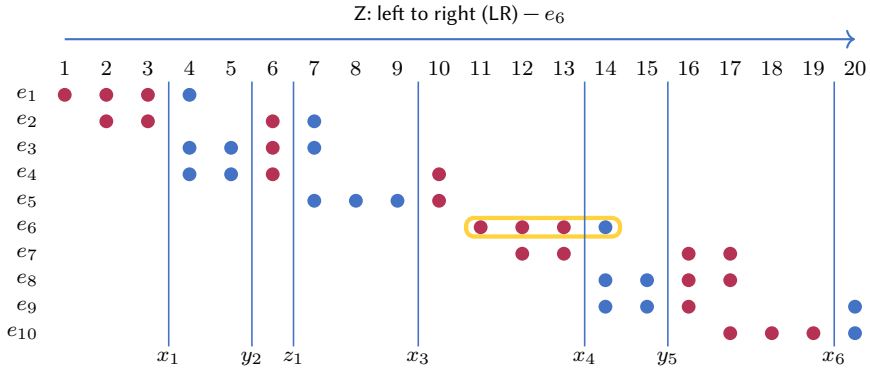


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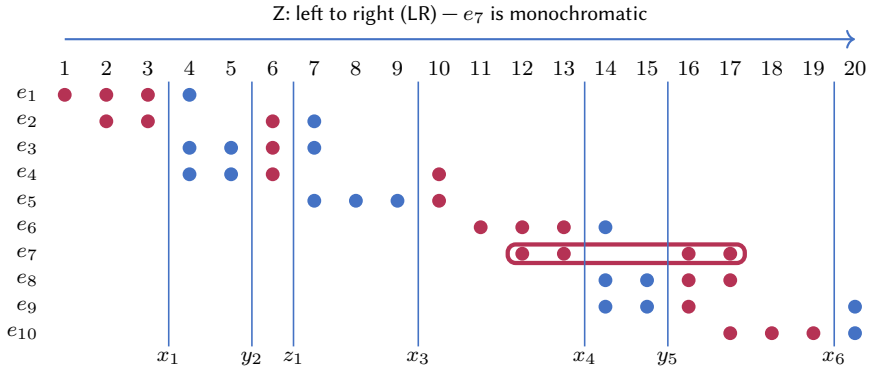


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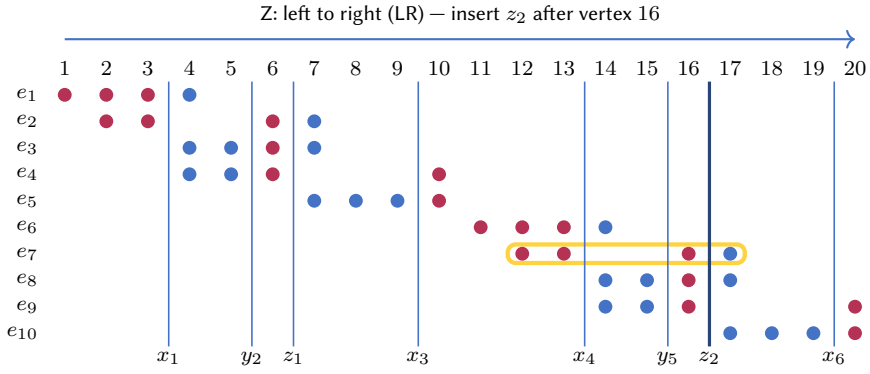


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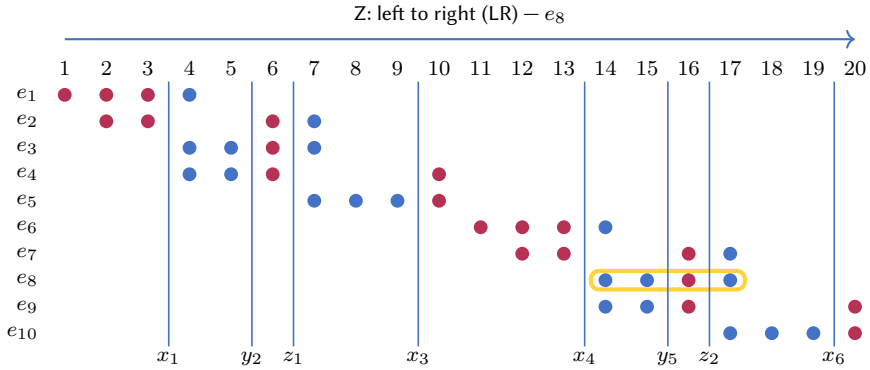


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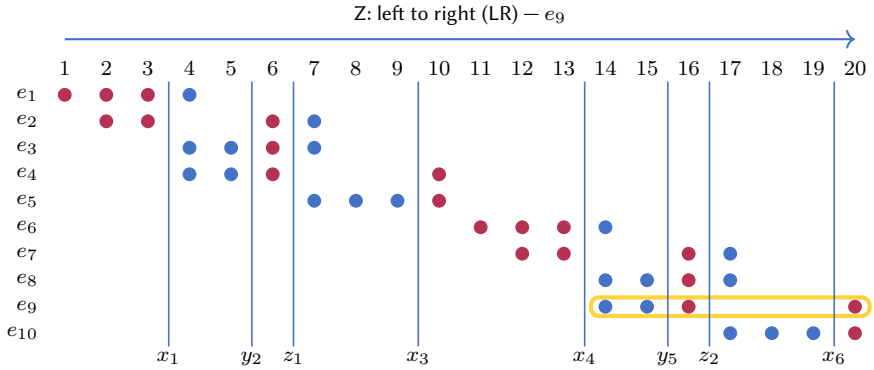


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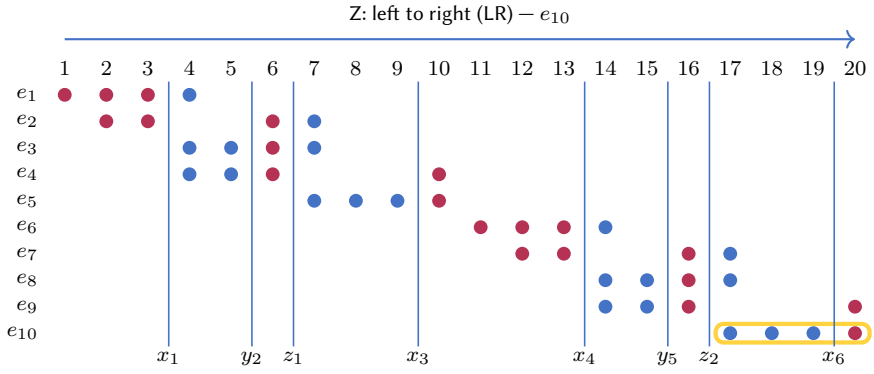


Algorithm outline

X: Colour greedily $\longrightarrow$; switch only to avoid a monochromatic edge with no switches.
X guarantees that at the end all monochromatic edges are (3,4) edges.

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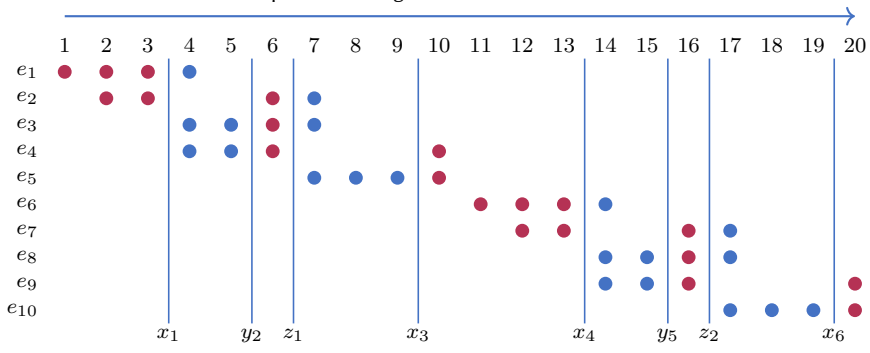


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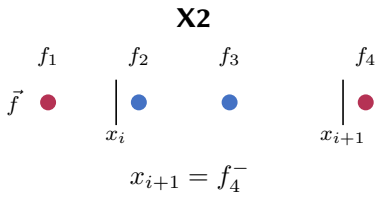


X1–X3: how the proof works

- **(X1)** Every edge $\vec{f}$ contains exactly one or two switches.

X1–X3: how the proof works

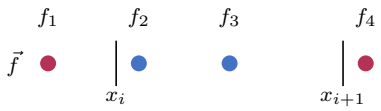
- **(X1)** Every edge $\vec{f}$ contains exactly one or two switches.
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X1–X3: how the proof works

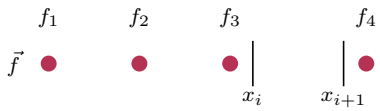
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- **(X3)** Every monochromatic edge in c_X is a $(3,4)$-edge.

X2



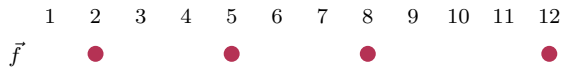
$x_{i+1} = f_4^-$

X3



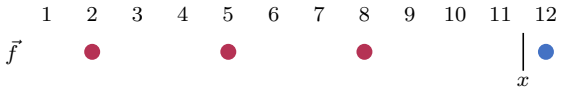
Both switches in (f_3, f_4).

X1: every edge contains at least one switch



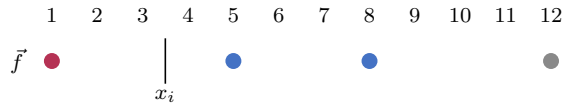
At least 1 switch by greedy colouring design – the greedy algorithm would always put a $x = f_4^-$ switch

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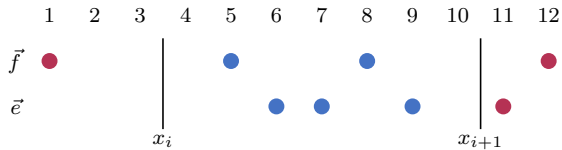
At least 1 switch by greedy colouring design – the greedy algorithm would always put a $x = f_4^-$ switch

X2: If there are at least two switches, the second is immediately before f_4



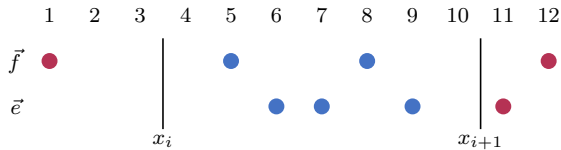
- The first switch x_i lies inside $\vec{f}$.

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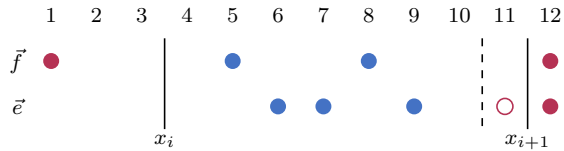
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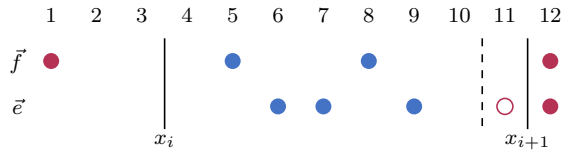
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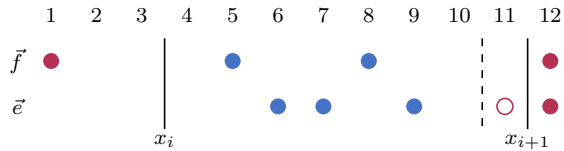
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X1. Every edge $\vec{f}$ has exactly one or two switches in c_X.
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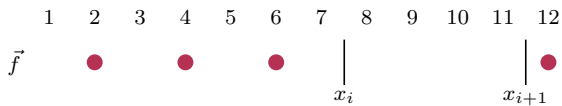
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Parity. A monochromatic edge has an even number of switches in each gap.

Claim

X3. *Every monochromatic edge in c_X is a $(3,4)$-edge.*

Proof

- *Let $\vec{f}$ be monochromatic in c_X.*



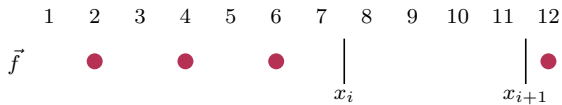
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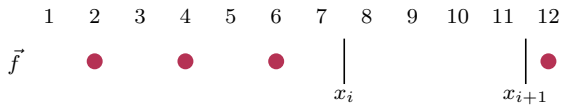
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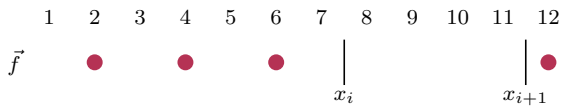
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- *Both switches lie in this gap, so $\vec{f}$ is a $(3,4)$-edge.*



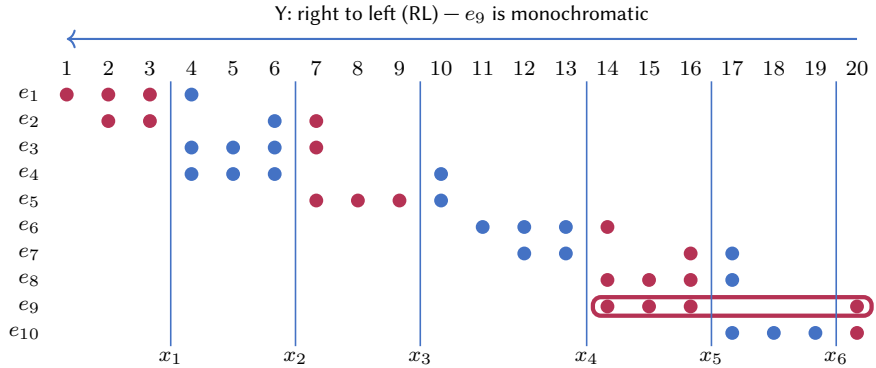
Stage X: conclusions

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Stage Y: recap

X3. Every monochromatic edge in c_X is a $(3,4)$-edge.

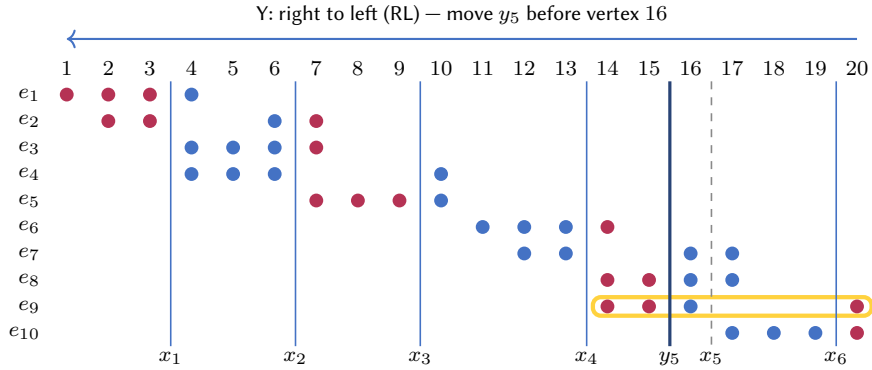
Sweep right to left: choose a $(3,4)$-edge $\vec{f}$ and move its first switch y_i just before f_3.



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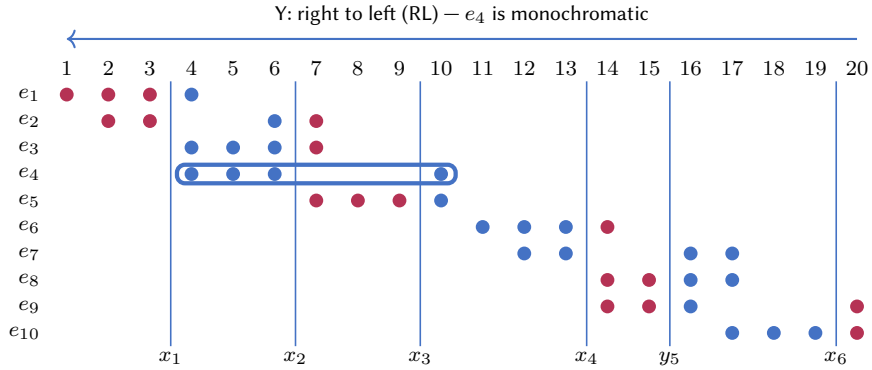
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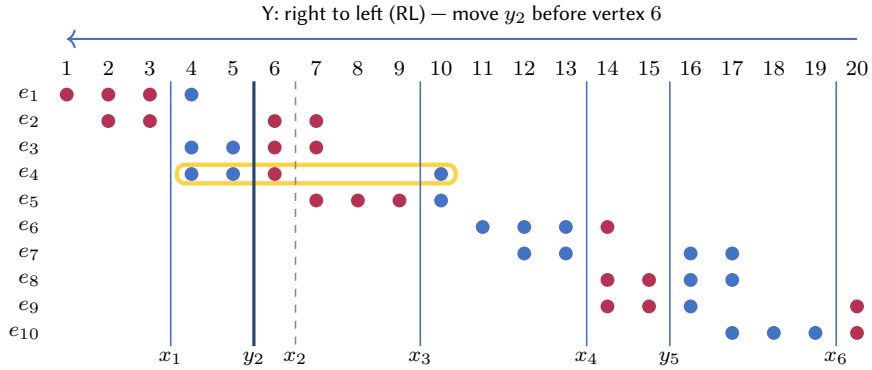
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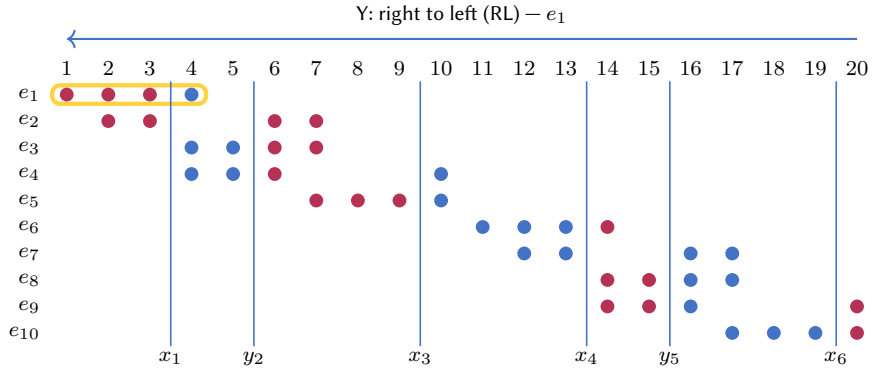
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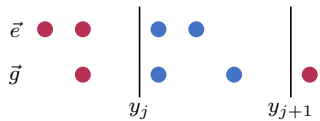


Y1–Y6: what Stage Y preserves and achieves

t: number of Stage Y switch moves performed
$c_{Y,t}$: colouring after t moves
$c_{Y,0} = c_X, \quad c_{Y,T} = c_Y$ (T: total number of moves)
At each step: moved edge $\vec{f}$, moved switch y_i; arbitrary edge $\vec{e}$.
Induction on t: Y1–Y5 preserved; Y6 at termination

- **Y1.** every edge has one or two switches in $c_{Y,t}$.

Y1: one or two switches



$\vec{e}$: one switch. $\vec{g}$: two switches.

Y1–Y6: what Stage Y preserves and achieves

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- **Y2.** $x_{j-1} < y_j(t) \leq x_j$.

Y2: no switch moves before the previous switch's original position



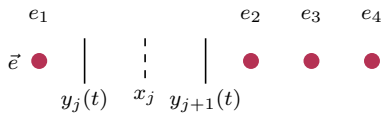
Dashed switches: original positions.

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- **Y1.** every edge has one or two switches in $c_{Y,t}$.
- **Y2.** $x_{j-1} < y_j(t) \leq x_j$.
- **Y3.** For $\vec{e} \in E(H), e_1 < x_j \Rightarrow e_2 < y_{j+1}(t)$.

Y3: forbidden counterexample

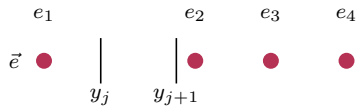


$e_1 < x_j$ but $y_{j+1}(t) < e_2$: impossible.

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- **Y4.** There are no $(1,2)$-edges in $c_{Y,t}$.



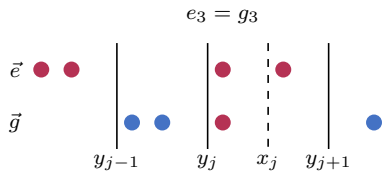
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- **Y4.** There are no $(1,2)$-edges in $c_{Y,t}$.
- **Y5.** If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.

Y5: a witness for a $(2,3)$-edge



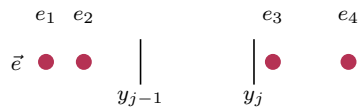
$e_3^- = y_j = g_3^- < x_j$
$y_{j+1} < g_4$.

Y1–Y6: what Stage Y preserves and achieves

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- **Y6.** Every monochromatic edge in c_Y is a $(2,3)$-edge.

Y6: the final monochromatic pattern



In c_Y: both switches in (e_2, e_3).

Y1–Y5: one Stage Y step

Induction on t: assume Y1–Y5 at time t.

Choose a maximal $(3,4)$-edge $\vec{f}$. Move its first switch $y_i(t)$, setting $y_i(t+1) = f_3^-$.

Let $\vec{e}$ denote an arbitrary edge.



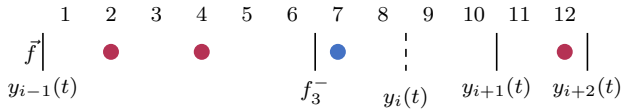
- $y_{i-1}(t) < f_1 < f_3 < y_i(t) < y_{i+1}(t) < f_4 < y_{i+2}(t)$.

Y1–Y5: one Stage Y step

Induction on t: assume Y1–Y5 at time t.

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- $y_{i-1}(t) < f_1 < f_3 < y_i(t) < y_{i+1}(t) < f_4 < y_{i+2}(t)$.
- $y_{i-1}(t) < f_1 < f_2 < y_i(t+1) = f_3^- < f_3 < y_{i+1}(t) < f_4$.

Auxillary claim: the moved edge starts after x_{i-1}

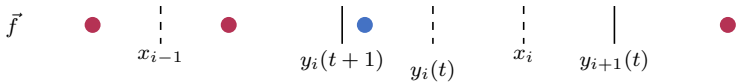
Y2. $x_{j-1} < y_j(t) \leq x_j$.
X2. If $\vec{e}$ contains x_j, x_{j+1}, then $x_{j+1} = e_4^-$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

For the moved edge $\vec{f}$ and its moved switch y_i, $x_{i-1} < f_1$.

Proof

- *Suppose $f_1 < x_{i-1}$.*



Auxillary claim: the moved edge starts after x_{i-1}

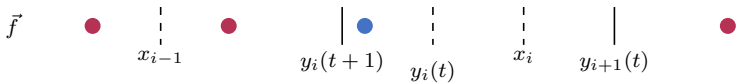
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X2. If $\vec{e}$ contains x_j, x_{j+1}, then $x_{j+1} = e_4^-$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

For the moved edge $\vec{f}$ and its moved switch y_i, $x_{i-1} < f_1$.

Proof

- *Suppose $f_1 < x_{i-1}$.*
- *Y2 gives $x_i < y_{i+1}(t) < f_4$, so $\vec{f}$ contains x_{i-1}, x_i.*



Auxillary claim: the moved edge starts after x_{i-1}

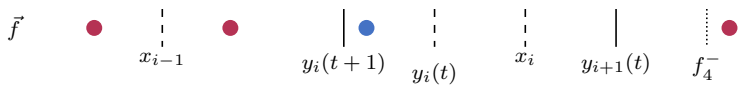
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Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

For the moved edge $\vec{f}$ and its moved switch y_i, $x_{i-1} < f_1$.

Proof

- *Suppose $f_1 < x_{i-1}$.*
- *Y2 gives $x_i < y_{i+1}(t) < f_4$, so $\vec{f}$ contains x_{i-1}, x_i.*
- *X2 forces $x_i = f_4^-$, contradicting $x_i < y_{i+1}(t) < f_4$.*



Y1: every edge keeps one or two switches

X1. Every edge $\vec{e}$ has exactly one or two switches in c_X.
X2. If $\vec{e}$ contains x_j, x_{j+1}, then $x_{j+1} = e_4^-$.
Y2. $x_{j-1} < y_j(t) \leq x_j$.
Auxiliary claim. $x_{i-1} < f_1$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim
Y1. *every edge has one or two switches in* $c_{Y,t}$.

Proof
- **Base.** $c_{Y,0} = c_X$ *: X1.*

Y1: every edge keeps one or two switches

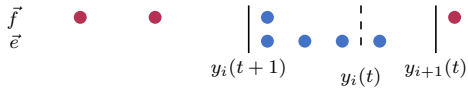
X1. Every edge $\vec{e}$ has exactly one or two switches in c_X.
X2. If $\vec{e}$ contains x_j, x_{j+1}, then $x_{j+1} = e_4^-$.
Y2. $x_{j-1} < y_j(t) \leq x_j$.
Auxiliary claim. $x_{i-1} < f_1$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

Y1. *every edge has one or two switches in $c_{Y,t}$.*

Proof

- **Base.** $c_{Y,0} = c_X$*: X1.*
- **Step: no zero.** *Losing the only switch gives $y_i(t+1) < e_1 < y_i(t) < e_4 < y_{i+1}(t)$. Thus $f_1 < e_1$ but $e_4 < f_4$: incomparable.*



Y1: every edge keeps one or two switches

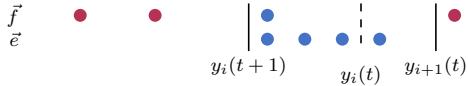
X1. Every edge $\vec{e}$ has exactly one or two switches in c_X.
X2. If $\vec{e}$ contains x_j, x_{j+1}, then $x_{j+1} = e_4^-$.
Y2. $x_{j-1} < y_j(t) \le x_j$.
Auxiliary claim. $x_{i-1} < f_1$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

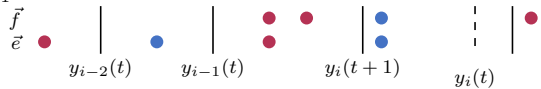
Y1. *every edge has one or two switches in* $c_{Y,t}$.

Proof

- **Base.** $c_{Y,0} = c_X$*: X1.*
- **Step: no zero.** *Losing the only switch gives* $y_i(t+1) < e_1 < y_i(t) < e_4 < y_{i+1}(t)$. *Thus* $f_1 < e_1$ *but* $e_4 < f_4$*: incomparable.*



- **No third switch.** *A new third switch* $y_i(t+1) = f_3^-$ *joins* $y_{i-2}(t), y_{i-1}(t)$. *By Y2,* $e_1 < y_{i-2}(t) \le x_{i-2} < y_{i-1}(t) \le x_{i-1}$.



Y1: every edge keeps one or two switches

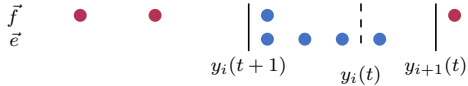
X1. Every edge $\vec{e}$ has exactly one or two switches in c_X.
X2. If $\vec{e}$ contains x_j, x_{j+1}, then $x_{j+1} = e_4^-$.
Y2. $x_{j-1} < y_j(t) \le x_j$.
Auxiliary claim. $x_{i-1} < f_1$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

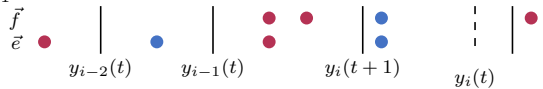
Y1. *every edge has one or two switches in* $c_{Y,t}$.

Proof

- **Base.** $c_{Y,0} = c_X$*: X1.*
- **Step: no zero.** *Losing the only switch gives* $y_i(t+1) < e_1 < y_i(t) < e_4 < y_{i+1}(t)$*. Thus* $f_1 < e_1$ *but* $e_4 < f_4$*: incomparable.*



- **No third switch.** *A new third switch* $y_i(t+1) = f_3^-$ *joins* $y_{i-2}(t), y_{i-1}(t)$*. By Y2,* $e_1 < y_{i-2}(t) \le x_{i-2} < y_{i-1}(t) \le x_{i-1}$.



- *Since* $x_{i-1} < f_1 < f_3 \le e_4$*, X2 gives* $x_{i-1} = e_4^- > f_1$*: a contradiction.*

Y2: no switch moves before the previous switch's original position

Auxiliary claim. $x_{i-1} < f_1$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim
Y2. $x_{j-1} < y_j(t) \leq x_j$.

Proof
- **Base.** $y_j(0) = x_j$.

Y2: no switch moves before the previous switch's original position



Dashed switches: original positions.

Y2: no switch moves before the previous switch's original position

Auxiliary claim. $x_{i-1} < f_1$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim
Y2. $x_{j-1} < y_j(t) \leq x_j$.

Proof

- **Base.** $y_j(0) = x_j$.
- **Step.** *Only $y_i(t)$ moves, every other switch keeps it position:*

Y2: no switch moves before the previous switch's original position



Dashed switches: original positions.

Y2: no switch moves before the previous switch's original position

Auxiliary claim. $x_{i-1} < f_1$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

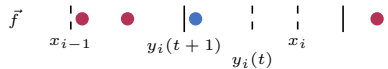
Claim

Y2. $x_{j-1} < y_j(t) \leq x_j$.

Proof

- **Base.** $y_j(0) = x_j$.
- **Step.** *Only $y_i(t)$ moves, every other switch keeps it position:*
- *For $y_i(t)$ we have:*

$$x_{i-1} < f_1 < f_3^- < y_i(t) \leq x_i.$$



$$x_{i-1} < y_i(t+1) < y_i(t) \leq x_i.$$

Y2: no switch moves before the previous switch's original position



Dashed switches: original positions.

Y3: the next switch stays past e_2

X2. If $\vec{e}$ contains x_j, x_{j+1}, then $x_{j+1} = e_4^-$.
Auxiliary claim. $x_{i-1} < f_1$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

Y3. *For $\vec{e} \in E(H)$, $e_1 < x_j \Rightarrow e_2 < y_{j+1}(t)$.*

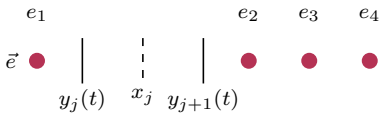
Proof

- **Base.** *If $e_1 < x_j < x_{j+1} \leq e_2$, then $\vec{e}$ contains both original switches.*



X2 requires $x_{j+1} = e_4^- > e_2$.

Y3: forbidden counterexample



$e_1 < x_j$ but $y_{j+1}(t) < e_2$: impossible.

Y3: the next switch stays past e_2

X2. If $\vec{e}$ contains x_j, x_{j+1}, then $x_{j+1} = e_4^-$.
Auxiliary claim. $x_{i-1} < f_1$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

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Y3. *For $\vec{e} \in E(H)$, $e_1 < x_j \Rightarrow e_2 < y_{j+1}(t)$.*

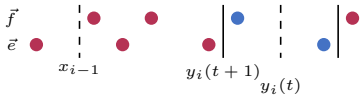
Proof

- **Base.** *If $e_1 < x_j < x_{j+1} \le e_2$, then $\vec{e}$ contains both original switches.*



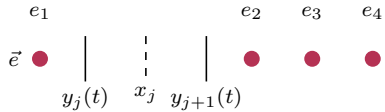
X2 requires $x_{j+1} = e_4^- > e_2$.

- **Step.** *Only $j = i - 1$ changes. If $e_1 < x_{i-1} < f_1$, comparability gives $\vec{e} < \vec{f}$.*



$e_2 < f_2 < f_1^- = y_i(t+1)$

Y3: forbidden counterexample



$e_1 < x_j$ but $y_{j+1}(t) < e_2$: impossible.

Y4: no $(1, 2)$-edge appears

X3. Every monochromatic edge in c_X is a $(3, 4)$-edge.
Y1. every edge has one or two switches in $c_{Y,t}$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

Y4. *There are no* $(1, 2)$*-edges in* $c_{Y,t}$.

Proof

- **Base.** *X3 excludes* $(1, 2)$*-edges in* $c_{Y,0}$.

Y4: no $(1, 2)$-edge appears

X3. Every monochromatic edge in c_X is a $(3, 4)$-edge.
Y1. every edge has one or two switches in $c_{Y,t}$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim
Y4. *There are no* $(1, 2)$*-edges in* $c_{Y,t}$.

Proof

- **Base.** *X3 excludes* $(1, 2)$*-edges in* $c_{Y,0}$.
- **Step.** *Suppose* $e_1 < y_j(t + 1) < y_{j+1}(t + 1) < e_2$.

Y4: no $(1, 2)$-edge appears

X3. Every monochromatic edge in c_X is a $(3, 4)$-edge.
Y1. every edge has one or two switches in $c_{Y,t}$.
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Claim

Y4. *There are no* $(1, 2)$*-edges in* $c_{Y,t}$.

Proof

- **Base.** *X3 excludes* $(1, 2)$*-edges in* $c_{Y,0}$.
- **Step.** *Suppose* $e_1 < y_j(t + 1) < y_{j+1}(t + 1) < e_2$.
- *Case when none of the switches is the moved switch trivial.*

Y4: no $(1,2)$-edge appears

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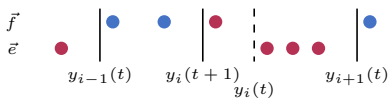
Claim
Y4. *There are no $(1,2)$-edges in $c_{Y,t}$.*

Proof

- **Base.** *X3 excludes $(1,2)$-edges in $c_{Y,0}$.*
- **Step.** *Suppose $e_1 < y_j(t+1) < y_{j+1}(t+1) < e_2$.*
- *Case when none of the switches is the moved switch trivial.*
- *If $j = i-1$,*
  $e_1 < y_{i-1}(t) < f_1 < f_2 < y_i(t+1) = f_3^- < e_2$.

  So $e_1 < f_1$ but $f_2 < e_2$: incomparable.

Case $j = i-1$



$e_1 < f_1$, but $f_2 < e_2$.

Y4: no $(1,2)$-edge appears

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Y1. every edge has one or two switches in $c_{Y,t}$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

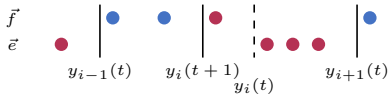
Claim

Y4. *There are no $(1,2)$-edges in $c_{Y,t}$.*

Proof

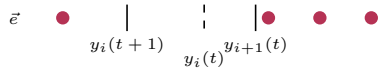
- **Base.** *X3 excludes $(1,2)$-edges in $c_{Y,0}$.*
- **Step.** *Suppose $e_1 < y_j(t+1) < y_{j+1}(t+1) < e_2$.*
- *Case when none of the switches is the moved switch trivial.*
- *If $j = i-1$,*
  $$e_1 < y_{i-1}(t) < f_1 < f_2 < y_i(t+1) = f_3^- < e_2.$$
  So $e_1 < f_1$ but $f_2 < e_2$: incomparable.
- *If $j = i$,*
  $$e_1 < y_i(t+1) < y_i(t) < y_{i+1}(t) < e_2.$$
  By Y1, a $(1,2)$-edge already at time t: contrary to induction.

Case $j = i-1$



$e_1 < f_1$, but $f_2 < e_2$.

Case $j = i$



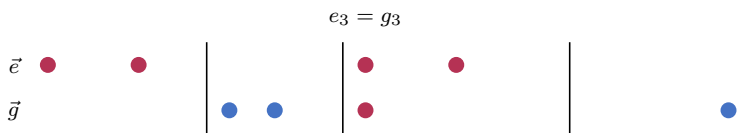
Both old switches were in (e_1, e_2).

Y5: unchanged switches keep their witness

X3. Every monochromatic edge in c_X is a $(3,4)$-edge.
Y1. every edge has one or two switches in $c_{Y,t}$.
Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim
Y5 holds at $t = 0$ and is preserved when neither switch of $\vec{e}$ moves.

Proof
- **Base.** *X3: no $(2,3)$-edges in $c_{Y,0}$.*



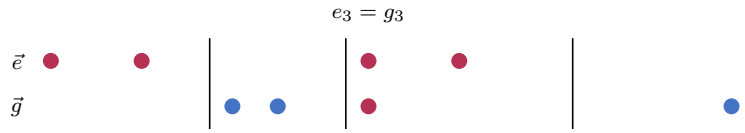
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Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim
Y5 holds at $t = 0$ and is preserved when neither switch of $\vec{e}$ moves.

Proof

- **Base.** *X3: no $(2,3)$-edges in $c_{Y,0}$.*
- *After the move, $\vec{f}$ is not monochromatic (only f_3 changes colour), so $\vec{e} \neq \vec{f}$.*



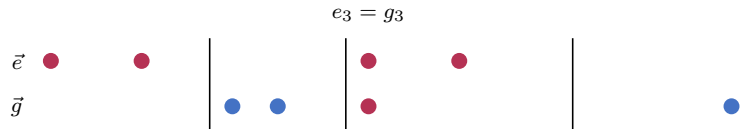
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Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim
Y5 holds at $t = 0$ and is preserved when neither switch of $\vec{e}$ moves.

Proof

- **Base.** *X3: no $(2,3)$-edges in $c_{Y,0}$.*
- *After the move, $\vec{f}$ is not monochromatic (only f_3 changes colour), so $\vec{e} \neq \vec{f}$.*
- *Let $\vec{e}$ be a $(2,3)$-edge at $t+1$ with both switches fixed. By Y1, it was already a $(2,3)$-edge at t.*



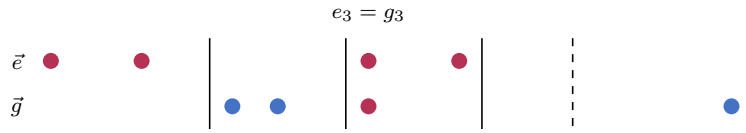
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Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim
Y5 holds at $t = 0$ and is preserved when neither switch of $\vec{e}$ moves.

Proof

- **Base.** *X3: no $(2,3)$-edges in $c_{Y,0}$.*
- *After the move, $\vec{f}$ is not monochromatic (only f_3 changes colour), so $\vec{e} \neq \vec{f}$.*
- *Let $\vec{e}$ be a $(2,3)$-edge at $t+1$ with both switches fixed. By Y1, it was already a $(2,3)$-edge at t.*
- *The next switch ($y_{j+1}(t)$ may move left, the same witness still works.*



Y5: moving the right switch gives witness $\vec{f}$

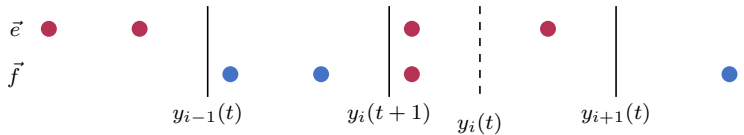
Y1. every edge has one or two switches in $c_{Y,t}$.
Y2. $x_{j-1} < y_j(t) \leq x_j$.
Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

If the $(2,3)$-edge $\vec{e}$ has switches $y_{i-1}(t), y_i(t+1)$, then $e_3 = f_3$ and $\vec{g} = \vec{f}$ is a witness.

Proof

- *Y1 puts the next switch outside $\vec{e}$:* $e_4 < y_{i+1}(t) < f_4$.



Y5: moving the right switch gives witness $\vec{f}$

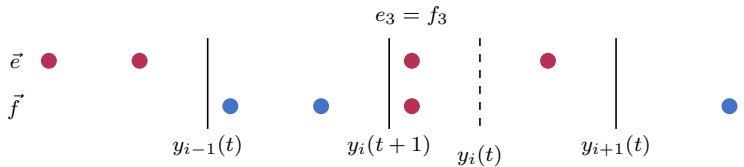
Y1. every edge has one or two switches in $c_{Y,t}$.
Y2. $x_{j-1} < y_j(t) \leq x_j$.
Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

If the $(2,3)$-edge $\vec{e}$ has switches $y_{i-1}(t), y_i(t+1)$, then $e_3 = f_3$ and $\vec{g} = \vec{f}$ is a witness.

Proof

- *Y1 puts the next switch outside $\vec{e}$: $e_4 < y_{i+1}(t) < f_4$.*
- *Comparability gives $e_3 \leq f_3$; $f_3^- = y_i(t+1) < e_3$ gives $f_3 \leq e_3$. Thus $e_3 = f_3$.*



Y5: moving the right switch gives witness $\vec{f}$

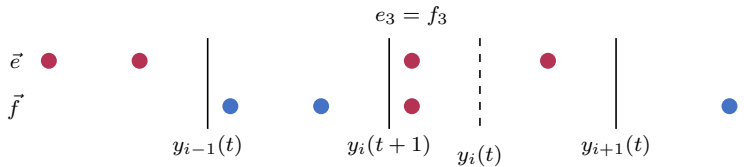
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Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

If the $(2,3)$-edge $\vec{e}$ has switches $y_{i-1}(t), y_i(t+1)$, then $e_3 = f_3$ and $\vec{g} = \vec{f}$ is a witness.

Proof

- *Y1 puts the next switch outside $\vec{e}$: $e_4 < y_{i+1}(t) < f_4$.*
- *Comparability gives $e_3 \leq f_3$; $f_3^- = y_i(t+1) < e_3$ gives $f_3 \leq e_3$. Thus $e_3 = f_3$.*
- *Take $\vec{g} = \vec{f}$: $e_3^- = y_i(t+1) = g_3^- < y_i(t) \leq x_i$, and $y_{i+1}(t+1) = y_{i+1}(t) < g_4$.*

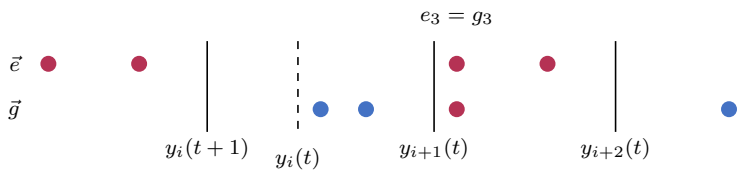


Y5: moving the left switch keeps the old witness

Y1. every edge has one or two switches in $c_{Y,t}$.
Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim
If the $(2,3)$-edge $\vec{e}$ has switches $y_i(t+1), y_{i+1}(t)$, its old witness still works.

Proof
- *Both old switches were already in the middle gap:* $e_2 < y_i(t+1) < y_i(t) < y_{i+1}(t) < e_3$.



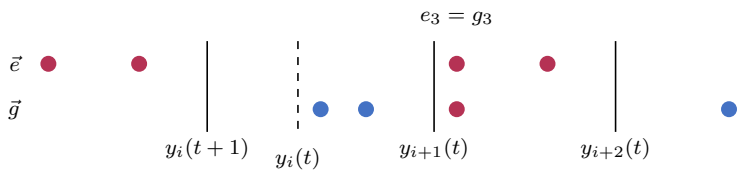
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If the $(2,3)$-edge $\vec{e}$ has switches $y_i(t+1), y_{i+1}(t)$, its old witness still works.

Proof

- *Both old switches were already in the middle gap: $e_2 < y_i(t+1) < y_i(t) < y_{i+1}(t) < e_3$.*
- *Y1 excludes any further switch at t; $\vec{e}$ was a $(2,3)$-edge, so induction gives $\vec{g}$.*



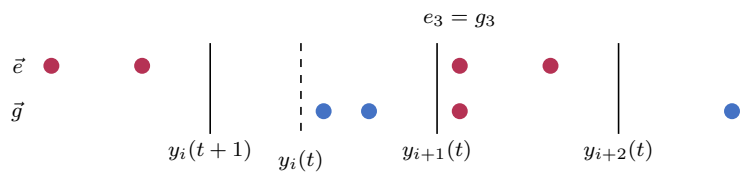
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Proof

- *Both old switches were already in the middle gap: $e_2 < y_i(t+1) < y_i(t) < y_{i+1}(t) < e_3$.*
- *Y1 excludes any further switch at t; $\vec{e}$ was a $(2,3)$-edge, so induction gives $\vec{g}$.*
- *The right switch and the next switch stay fixed: $e_3^- = y_{i+1}(t+1) = g_3^- < x_{i+1}$, and $y_{i+2}(t+1) = y_{i+2}(t) < g_4$.*



Y6: chosen $(3, 4)$-edges move left

Y1. every edge has one or two switches in $c_{Y,t}$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

$\vec{f}(t+1) < \vec{f}(t)$.

Proof

- *Suppose the next chosen edge $\vec{e} > \vec{f}$. It must be new, since $\vec{f}$ was maximal.*

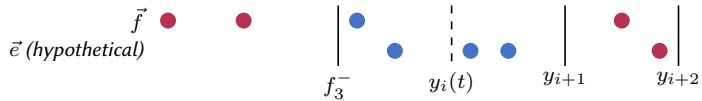
Y6: chosen $(3,4)$-edges move left

Y1. every edge has one or two switches in $c_{Y,t}$.
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Claim
$\vec{f}(t+1) < \vec{f}(t)$.

Proof

- *Suppose the next chosen edge $\vec{e} > \vec{f}$. It must be new, since $\vec{f}$ was maximal.*
- $y_i(t+1) = f_3^- < f_3 \leq e_3$. *A $(3,4)$-edge has both switches after e_3, so the new switch is outside $\vec{e}$.*



The moved switch lies left of e_1: it cannot supply a new pair in (e_3, e_4).

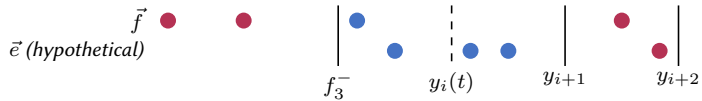
Y6: chosen $(3,4)$-edges move left

Y1. every edge has one or two switches in $c_{Y,t}$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim
$\vec{f}(t+1) < \vec{f}(t)$.

Proof

- *Suppose the next chosen edge $\vec{e} > \vec{f}$. It must be new, since $\vec{f}$ was maximal.*
- *$y_i(t+1) = f_3^- < f_3 \le e_3$. A $(3,4)$-edge has both switches after e_3, so the new switch is outside $\vec{e}$.*
- *If the old switch was outside too, $\vec{e}$ was already monochromatic. If inside, Y1 leaves at most one switch. Both are impossible.*



The moved switch lies left of e_1: it cannot supply a new pair in (e_3, e_4).

Y6: termination leaves only $(2,3)$-edges

Y1. every edge has one or two switches in $c_{Y,t}$.
Y4. There are no $(1,2)$-edges in $c_{Y,t}$.
Previous step. $\vec{f}(t+1) < \vec{f}(t)$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

Y6. *Every monochromatic edge in c_Y is a $(2,3)$-edge.*

Proof

- *By the previous step, the chosen edges strictly decrease, so Stage Y terminates.*

Y6: termination leaves only $(2,3)$-edges

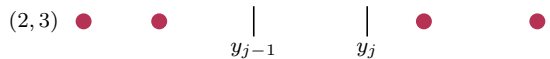
Y1. every edge has one or two switches in $c_{Y,t}$.
Y4. There are no $(1,2)$-edges in $c_{Y,t}$.
Previous step. $\vec{f}(t+1) < \vec{f}(t)$.
Notation. Moved edge $\vec{f}$; moved switch y_i; arbitrary edge $\vec{e}$.

Claim

Y6. *Every monochromatic edge in c_Y is a $(2,3)$-edge.*

Proof

- *By the previous step, the chosen edges strictly decrease, so Stage Y terminates.*
- *At termination: no $(3,4)$-edge; Y4 excludes $(1,2)$-edges. By Y1, only $(2,3)$-edges can be monochromatic.*

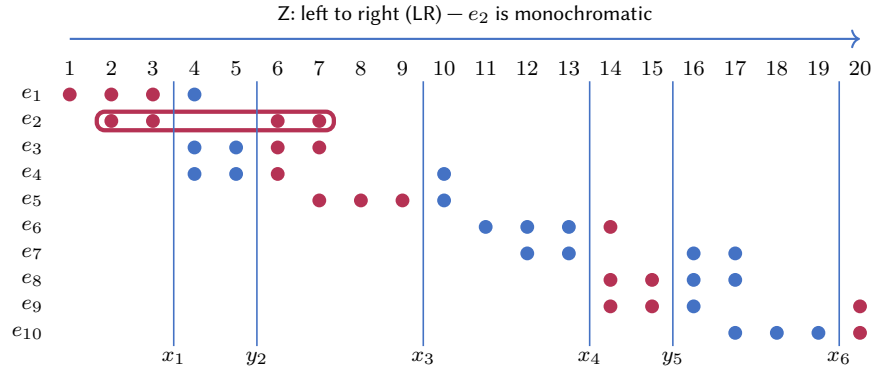


The only surviving monochromatic pattern: two switches in the middle gap.

Stage Z: recap

Y6. Every monochromatic edge in c_Y is a $(2,3)$-edge.

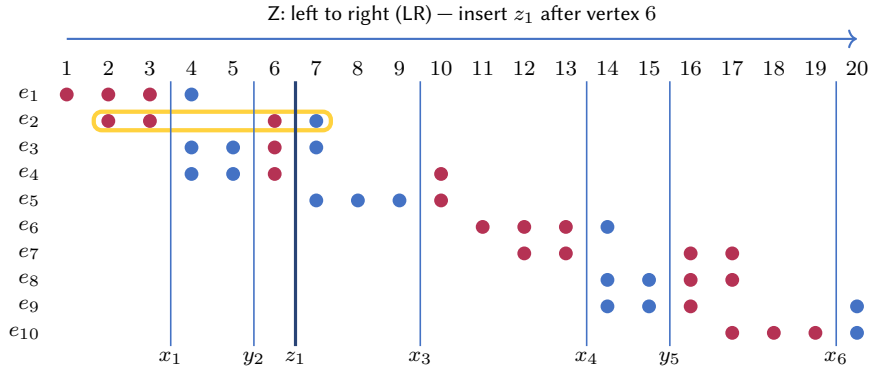
Sweep left to right; insert a switch just after the third vertex of each remaining $(2,3)$-edge.



Stage Z: recap

Y6. Every monochromatic edge in c_Y is a $(2,3)$-edge.

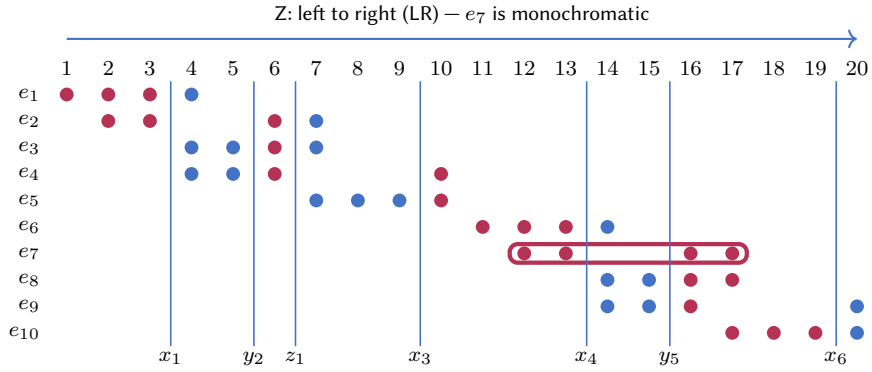
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Stage Z: recap

Y6. Every monochromatic edge in c_Y is a $(2,3)$-edge.

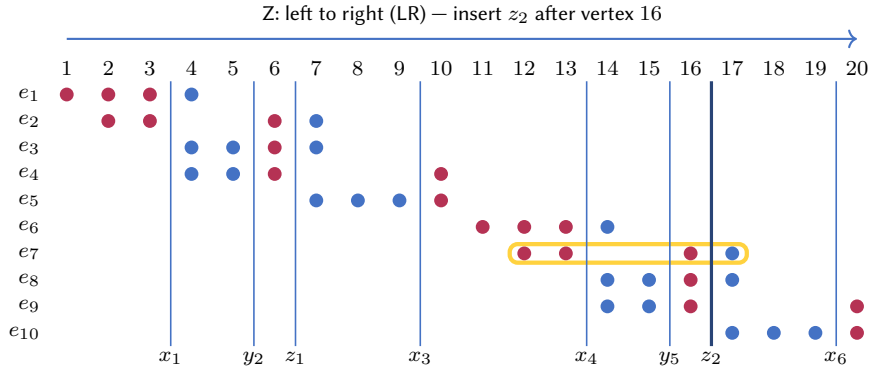
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Stage Z: recap

Y6. Every monochromatic edge in c_Y is a $(2,3)$-edge.

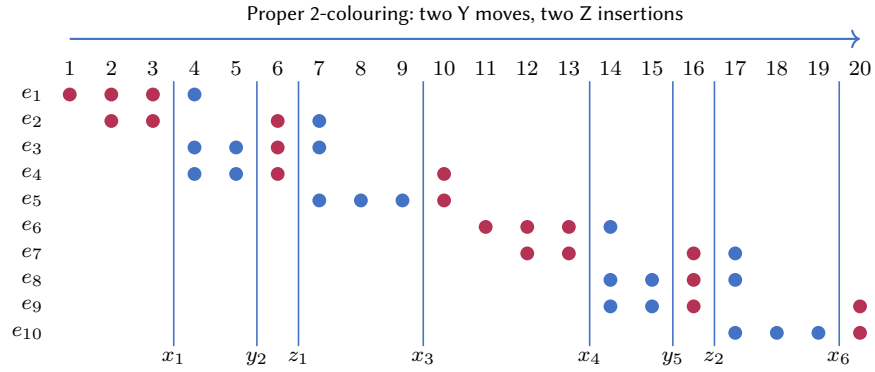
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Stage Z: recap

Y6. Every monochromatic edge in c_Y is a $(2,3)$-edge.

Sweep left to right; insert a switch just after the third vertex of each remaining $(2,3)$-edge.

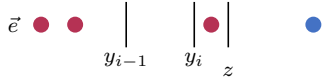


Z1–Z4: what Stage Z preserves and achieves

t: number of Stage Z insertions; $Z(t)$: inserted switches
$c_{Z,t}$: colouring with switches $Y \cup Z(t)$; $c_{Z,0} = c_Y$
$M(c)$: monochromatic edges in c
Induction on t: Z1–Z4; the Y-switches stay fixed

- **Z1.** $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.

Z1: at most three switches



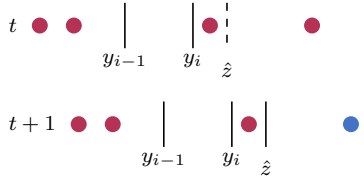
Two old switches, one new switch.

Z1–Z4: what Stage Z preserves and achieves

t: number of Stage Z insertions; $Z(t)$: inserted switches
$c_{Z,t}$: colouring with switches $Y \cup Z(t)$; $c_{Z,0} = c_Y$
$M(c)$: monochromatic edges in c
Induction on t: Z1–Z4; the Y-switches stay fixed

- **Z1.** $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
- **Z2.** $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.

Z2: repair the chosen edge



No new monochromatic edges.

Z1–Z4: what Stage Z preserves and achieves

t: number of Stage Z insertions; $Z(t)$: inserted switches
$c_{Z,t}$: colouring with switches $Y \cup Z(t)$; $c_{Z,0} = c_Y$
$M(c)$: monochromatic edges in c
Induction on t: Z1–Z4; the Y-switches stay fixed

- **Z1.** $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
- **Z2.** $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
- **Z3.** Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.

Z3: surviving monochromatic edges



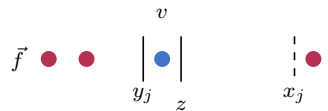
$z < f_1$ for every $z \in Z(t)$.

Z1–Z4: what Stage Z preserves and achieves

t: number of Stage Z insertions; $Z(t)$: inserted switches
$c_{Z,t}$: colouring with switches $Y \cup Z(t)$; $c_{Z,0} = c_Y$
$M(c)$: monochromatic edges in c
Induction on t: Z1–Z4; the Y-switches stay fixed

- **Z1.** $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
- **Z2.** $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
- **Z3.** Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
- **Z4.** For each $z \in Z(t)$, some $j \in [s]$ and $v \in [n]$ satisfy $y_j = v^-$ and $z = v^+ \leq x_j$.

Z4: two switches bracket a vertex

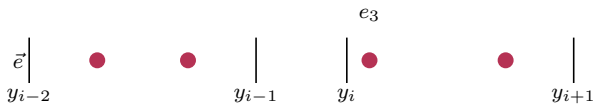


An inserted z occurs only immediately right of v:
$y_j = v^-, \quad z = v^+ \leq x_j.$

Z1–Z4: one Stage Z step

Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.

Assume Z1–Z4 at time t; choose a minimal monochromatic edge $\vec{e}$. Insert $\hat{z} = e_3^+$.

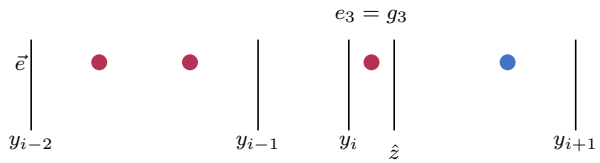


- Z3: $\vec{e}$ has two Y-switches in (e_2, e_3); every old $z \in Z(t)$ lies before e_1.

Z1–Z4: one Stage Z step

Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.

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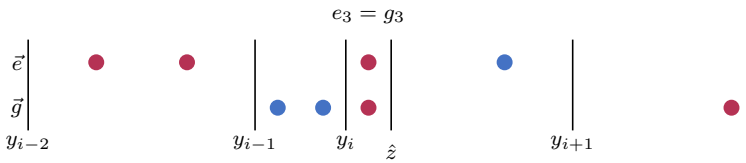


- Z3: $\vec{e}$ has two Y-switches in (e_2, e_3); every old $z \in Z(t)$ lies before e_1.
- Y5: $y_i = e_3^- < x_i$; inserting $\hat{z} = e_3^+$ changes the colour of e_4.

Z1–Z4: one Stage Z step

Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.

Assume Z1–Z4 at time t; choose a minimal monochromatic edge $\vec{e}$. Insert $\hat{z} = e_3^+$.



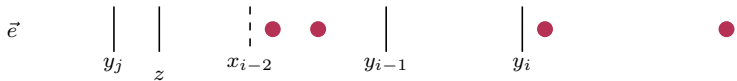
- Z3: $\vec{e}$ has two Y-switches in (e_2, e_3); every old $z \in Z(t)$ lies before e_1.
- Y5: $y_i = e_3^- < x_i$; inserting $\hat{z} = e_3^+$ changes the colour of e_4.
- Y5 also gives $g_3 = e_3$ and $y_{i+1} < g_4$.

Z1: older inserted switches lie before x_{i-2}

Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Z4. For each $z \in Z(t)$, some $j \in [s]$ and $v \in [n]$ satisfy $y_j = v^-$ and $z = v^+ \leq x_j$.
Y2. $x_{j-1} < y_j(t) \leq x_j$.

Claim
Every $z \in Z(t)$ satisfies $z \leq x_{i-2}$.

Proof
- *Z3 gives $z < e_1 < y_{i-1}$.*

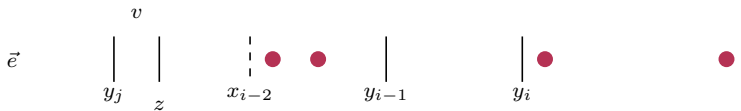


Z1: older inserted switches lie before x_{i-2}

Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Z4. For each $z \in Z(t)$, some $j \in [s]$ and $v \in [n]$ satisfy $y_j = v^-$ and $z = v^+ \leq x_j$.
Y2. $x_{j-1} < y_j(t) \leq x_j$.

Claim
Every $z \in Z(t)$ satisfies $z \leq x_{i-2}$.

Proof
- *Z3 gives $z < e_1 < y_{i-1}$.*
- *By Z4, $y_j = v^- < z = v^+ \leq x_j$ for some j.*



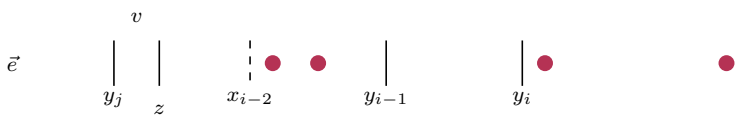
Z1: older inserted switches lie before x_{i-2}

Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Z4. For each $z \in Z(t)$, some $j \in [s]$ and $v \in [n]$ satisfy $y_j = v^-$ and $z = v^+ \leq x_j$.
Y2. $x_{j-1} < y_j(t) \leq x_j$.

Claim
Every $z \in Z(t)$ satisfies $z \leq x_{i-2}$.

Proof

- *Z3 gives $z < e_1 < y_{i-1}$.*
- *By Z4, $y_j = v^- < z = v^+ \leq x_j$ for some j.*
- *Thus $j \leq i-2$, so $z \leq x_j \leq x_{i-2}$. Y2 also gives $x_{i-2} < y_{i-1}$.*



Z1: every edge keeps at most three switches

Y1. every edge has one or two switches in $c_{Y,t}$.
Y2. $x_{j-1} < y_j(t) \leq x_j$.
X2. If $\vec{f}$ contains x_j, x_{j+1}, then $x_{j+1} = f_4^-$.
Z1. $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
Previous step. Every $z \in Z(t)$ satisfies $z \leq x_{i-2}$.

Claim
Z1. *$c_{Z,t}$ is 3-well-spread: every edge has at most three switches.*

Proof

- **Base.** *Y1: at most two switches in $c_{Z,0} = c_Y$.*

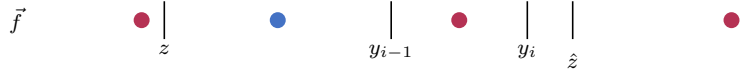
Z1: every edge keeps at most three switches

Y1. every edge has one or two switches in $c_{Y,t}$.
Y2. $x_{j-1} < y_j(t) \le x_j$.
X2. If $\vec{f}$ contains x_j, x_{j+1}, then $x_{j+1} = f_4^-$.
Z1. $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
Previous step. Every $z \in Z(t)$ satisfies $z \le x_{i-2}$.

Claim
Z1. *$c_{Z,t}$ is 3-well-spread: every edge has at most three switches.*

Proof

- **Base.** *Y1: at most two switches in $c_{Z,0} = c_Y$.*
- **Step.** *A new fourth switch in $\vec{f}$ requires both $\hat{z}$ and an old $z \in Z(t)$: Y1 allows only two Y-switches.*



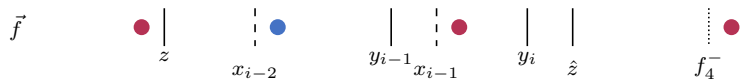
Z1: every edge keeps at most three switches

Y1. every edge has one or two switches in $c_{Y,t}$.
Y2. $x_{j-1} < y_j(t) \leq x_j$.
X2. If $\vec{f}$ contains x_j, x_{j+1}, then $x_{j+1} = f_4^-$.
Z1. $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
Previous step. Every $z \in Z(t)$ satisfies $z \leq x_{i-2}$.

Claim
Z1. *$c_{Z,t}$ is 3-well-spread: every edge has at most three switches.*

Proof

- **Base.** *Y1: at most two switches in $c_{Z,0} = c_Y$.*
- **Step.** *A new fourth switch in $\vec{f}$ requires both $\hat{z}$ and an old $z \in Z(t)$: Y1 allows only two Y-switches.*
- *By the previous step and Y2, $f_1 < z \leq x_{i-2} < x_{i-1} < y_i < \hat{z} < f_4$.*



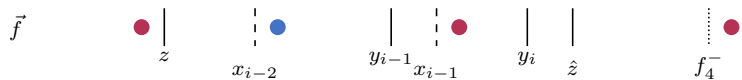
Z1: every edge keeps at most three switches

Y1. every edge has one or two switches in $c_{Y,t}$.
Y2. $x_{j-1} < y_j(t) \leq x_j$.
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Z1. $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
Previous step. Every $z \in Z(t)$ satisfies $z \leq x_{i-2}$.

Claim
Z1. *$c_{Z,t}$ is 3-well-spread: every edge has at most three switches.*

Proof

- **Base.** *Y1: at most two switches in $c_{Z,0} = c_Y$.*
- **Step.** *A new fourth switch in $\vec{f}$ requires both $\hat{z}$ and an old $z \in Z(t)$: Y1 allows only two Y-switches.*
- *By the previous step and Y2, $f_1 < z \leq x_{i-2} < x_{i-1} < y_i < \hat{z} < f_4$.*
- *X2 forces $x_{i-1} = f_4^-$, but $x_{i-1} < \hat{z} < f_4$.*



Z2: a new monochromatic edge has only two possible switch pairs

Y1. every edge has one or two switches in $c_{Y,t}$.
Z1. $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ *for* $t \geq 1$.

Proof
- **Base.** *No insertion at* $t = 0$*; Z2 starts at* $t = 1$.

Z2: a new monochromatic edge has only two possible switch pairs

Y1. every edge has one or two switches in $c_{Y,t}$.
Z1. $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ *for* $t \geq 1$.

Proof

- **Base.** *No insertion at* $t = 0$*; Z2 starts at* $t = 1$.
- **Step.** *The chosen* $\vec{e}$ *is no longer monochromatic. Suppose a new monochromatic edge* $\vec{f}$ *appears.*



Z2: a new monochromatic edge has only two possible switch pairs

Y1. every edge has one or two switches in $c_{Y,t}$.
Z1. $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ *for* $t \geq 1$.

Proof

- **Base.** *No insertion at* $t = 0$*; Z2 starts at* $t = 1$.
- **Step.** *The chosen* $\vec{e}$ *is no longer monochromatic. Suppose a new monochromatic edge* $\vec{f}$ *appears.*
- *It contains* $\hat{z}$*; otherwise all its vertices change colour together, or none do.*



Z2: a new monochromatic edge has only two possible switch pairs

Y1. every edge has one or two switches in $c_{Y,t}$.
Z1. $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ *for* $t \geq 1$.

Proof

- **Base.** *No insertion at* $t = 0$*; Z2 starts at* $t = 1$.
- **Step.** *The chosen* $\vec{e}$ *is no longer monochromatic. Suppose a new monochromatic edge* $\vec{f}$ *appears.*
- *It contains* $\hat{z}$*; otherwise all its vertices change colour together, or none do.*
- *Y1, Z1 and Parity: exactly two switches, in one gap.*



Z2: a new monochromatic edge has only two possible switch pairs

Y1. every edge has one or two switches in $c_{Y,t}$.
Z1. $c_{Z,t}$ is 3-well-spread: every edge has at most three switches.
Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ *for* $t \geq 1$.

Proof

- **Base.** *No insertion at* $t = 0$*; Z2 starts at* $t = 1$.
- **Step.** *The chosen* $\vec{e}$ *is no longer monochromatic. Suppose a new monochromatic edge* $\vec{f}$ *appears.*
- *It contains* $\hat{z}$*; otherwise all its vertices change colour together, or none do.*
- *Y1, Z1 and Parity: exactly two switches, in one gap.*
- *Z3 puts all old* z *before* e_1*. The partner of* $\hat{z}$ *is therefore* y_i *or* y_{i+1}.



Z2: $y_i, \hat{z}$ cannot lie in the last gap

Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim

No new monochromatic edge has only $y_i, \hat{z}$ in (f_3, f_4).

Proof

- *Suppose $f_3 < y_i < \hat{z} < f_4$.*



Z2: $y_i, \hat{z}$ cannot lie in the last gap

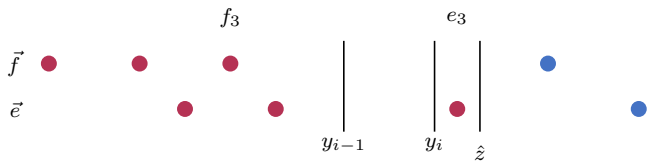
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim

No new monochromatic edge has only $y_i, \hat{z}$ in (f_3, f_4).

Proof

- *Suppose $f_3 < y_i < \hat{z} < f_4$.*
- *Since $f_3 < e_3$, comparability gives $\vec{f} < \vec{e}$ and $f_1 \leq e_1$.*



Z2: $y_i, \hat{z}$ cannot lie in the last gap

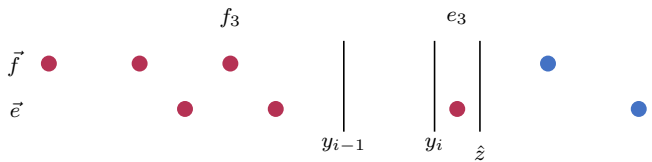
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim

No new monochromatic edge has only $y_i, \hat{z}$ in (f_3, f_4).

Proof

- *Suppose $f_3 < y_i < \hat{z} < f_4$.*
- *Since $f_3 < e_3$, comparability gives $\vec{f} < \vec{e}$ and $f_1 \leq e_1$.*
- *But $f_1 \leq e_1 < y_{i-1} < y_i < \hat{z} < f_4$: a third switch lies in $\vec{f}$.*



Z2: $y_i, \hat{z}$ cannot lie in either earlier gap

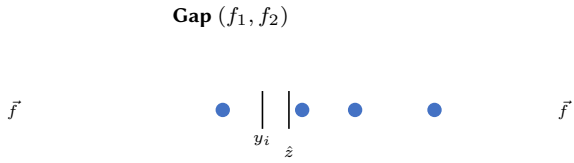
Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
Parity. A monochromatic edge has an even number of switches in each gap.

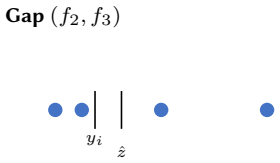
Claim

No new monochromatic edge has only $y_i, \hat{z}$ in (f_1, f_2) or (f_2, f_3).

Proof

- *In both cases, $\hat{z} < f_3$.*





Z2: $y_i, \hat{z}$ cannot lie in either earlier gap

Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
Parity. A monochromatic edge has an even number of switches in each gap.

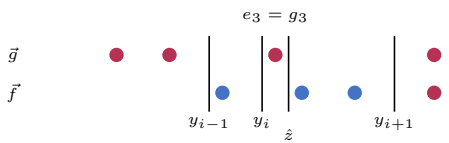
Claim

No new monochromatic edge has only $y_i, \hat{z}$ in (f_1, f_2) or (f_2, f_3).

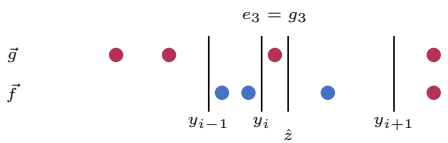
Proof

- *In both cases, $\hat{z} < f_3$.*
- *Y5 gives $g_3 = e_3 < \hat{z} < f_3$ and $y_{i+1} < g_4$.*

Gap (f_1, f_2)



Gap (f_2, f_3)



Z2: $y_i, \hat{z}$ cannot lie in either earlier gap

Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
Parity. A monochromatic edge has an even number of switches in each gap.

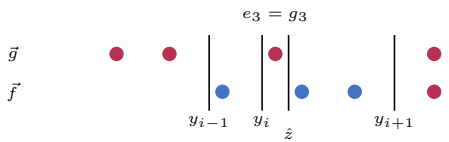
Claim

No new monochromatic edge has only $y_i, \hat{z}$ in (f_1, f_2) or (f_2, f_3).

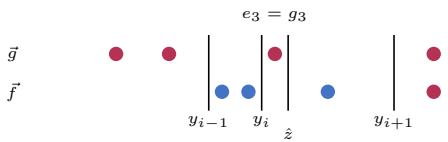
Proof

- *In both cases, $\hat{z} < f_3$.*
- *Y5 gives $g_3 = e_3 < \hat{z} < f_3$ and $y_{i+1} < g_4$.*
- *Comparability gives $\vec{g} < \vec{f}$, so $y_{i+1} < g_4 \leq f_4$: a third switch lies in $\vec{f}$.*

Gap (f_1, f_2)



Gap (f_2, f_3)



Z2: the pair $\hat{z}, y_{i+1}$ is impossible

Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Y3. For $\vec{e} \in E(H)$, $e_1 < x_j \Rightarrow e_2 < y_{j+1}(t)$.
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.

Claim

No new monochromatic edge has only $\hat{z}, y_{i+1}$.

Proof

- *y_i is outside $\vec{f}$, so $e_3^- = y_i < f_1 < \hat{z} = e_3^+$. Thus $f_1 = e_3$.*



Z2: the pair $\hat{z}, y_{i+1}$ is impossible

Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Y3. For $\vec{e} \in E(H)$, $e_1 < x_j \Rightarrow e_2 < y_{j+1}(t)$.
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.

Claim
No new monochromatic edge has only $\hat{z}, y_{i+1}$.

Proof

- *y_i is outside $\vec{f}$, so $e_3^- = y_i < f_1 < \hat{z} = e_3^+$. Thus $f_1 = e_3$.*
- *Both switches lie in the first gap: $y_{i+1} < f_2$.*



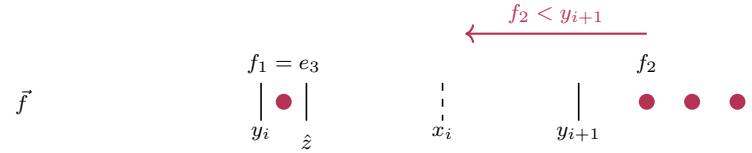
Z2: the pair $\hat{z}, y_{i+1}$ is impossible

Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Y3. For $\vec{e} \in E(H)$, $e_1 < x_j \Rightarrow e_2 < y_{j+1}(t)$.
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.

Claim
No new monochromatic edge has only $\hat{z}, y_{i+1}$.

Proof

- *y_i is outside $\vec{f}$, so $e_3^- = y_i < f_1 < \hat{z} = e_3^+$. Thus $f_1 = e_3$.*
- *Both switches lie in the first gap: $y_{i+1} < f_2$.*
- *Y5 gives $f_1 = e_3 < x_i$; Y3 gives $f_2 < y_{i+1}$: a contradiction.*



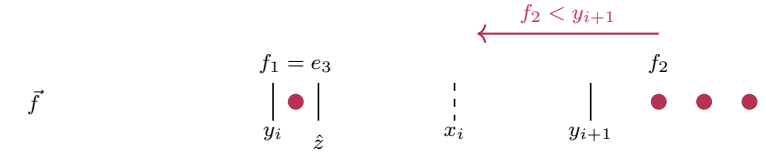
Z2: the pair $\hat{z}, y_{i+1}$ is impossible

Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Y3. For $\vec{e} \in E(H)$, $e_1 < x_j \Rightarrow e_2 < y_{j+1}(t)$.
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.

Claim
No new monochromatic edge has only $\hat{z}, y_{i+1}$.

Proof

- *y_i is outside $\vec{f}$, so $e_3^- = y_i < f_1 < \hat{z} = e_3^+$. Thus $f_1 = e_3$.*
- *Both switches lie in the first gap: $y_{i+1} < f_2$.*
- *Y5 gives $f_1 = e_3 < x_i$; Y3 gives $f_2 < y_{i+1}$: a contradiction.*
- *Both possible pairs are excluded. No new monochromatic edge appears, and $\vec{e}$ is repaired: Z2.*



Z3: surviving edges start after the new switch

Y6. Every monochromatic edge in c_Y is a $(2,3)$-edge.
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim

Z3. *Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.*

Proof

- **Base.** *Y6 gives the $(2,3)$ pattern; $Z(0) = \varnothing$.*

Z3: surviving edges start after the new switch

Y6. Every monochromatic edge in c_Y is a $(2,3)$-edge.
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim

Z3. *Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.*

Proof

- **Base.** *Y6 gives the $(2,3)$ pattern; $Z(0) = \varnothing$.*
- **Step.** *By Z2 and induction, a surviving $\vec{f}$ was a $(2,3)$-edge, and every old z lies before f_1.*



Z3: surviving edges start after the new switch

Y6. Every monochromatic edge in c_Y is a $(2,3)$-edge.
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim

Z3. *Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.*

Proof

- **Base.** *Y6 gives the $(2,3)$ pattern; $Z(0) = \varnothing$.*
- **Step.** *By Z2 and induction, a surviving $\vec{f}$ was a $(2,3)$-edge, and every old z lies before f_1.*
- *It cannot contain $\hat{z}$: its two old switches plus $\hat{z}$ violate Parity.*



Z3: surviving edges start after the new switch

Y6. Every monochromatic edge in c_Y is a $(2,3)$-edge.
Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.
Parity. A monochromatic edge has an even number of switches in each gap.

Claim

Z3. *Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.*

Proof

- **Base.** *Y6 gives the $(2,3)$ pattern; $Z(0) = \varnothing$.*
- **Step.** *By Z2 and induction, a surviving $\vec{f}$ was a $(2,3)$-edge, and every old z lies before f_1.*
- *It cannot contain $\hat{z}$: its two old switches plus $\hat{z}$ violate Parity.*
- *Minimality gives $\vec{e} < \vec{f}$, hence $\hat{z} < e_4 \leq f_4$. Thus $\hat{z} < f_1$; the two switches of $\vec{f}$ stay in (f_2, f_3).*



Z4: the new switch brackets the third vertex

Y5. If $\vec{e}$ is a $(2,3)$-edge with switches $y_{j-1}(t), y_j(t)$, then $e_3^- = y_j(t) < x_j$, and some $\vec{g} \in E(H)$ has $y_j(t) = g_3^-$ and $y_{j+1}(t) < g_4$.
Z4. For each $z \in Z(t)$, some $j \in [s]$ and $v \in [n]$ satisfy $y_j = v^-$ and $z = v^+ \le x_j$.

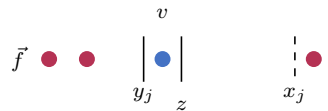
Claim

Z4. *For each $z \in Z(t)$, some $j \in [s]$ and $v \in [n]$ satisfy $y_j = v^-$ and $z = v^+ \le x_j$.*

Proof

- **Base.** $Z(0) = \varnothing$.

Z4: two switches bracket a vertex



An inserted z occurs only immediately right of v:

$$y_j = v^-, \quad z = v^+ \le x_j.$$

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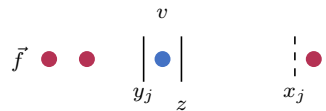
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Proof

- **Base.** $Z(0) = \varnothing$.
- **Step.** *Every older inserted switch keeps its witness: the Y-switches stay fixed.*

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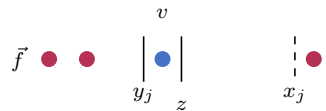
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- **Step.** *Every older inserted switch keeps its witness: the Y-switches stay fixed.*
- *Y5 gives $y_i = e_3^- < x_i$. The next possible switch position is e_3^+, so $\hat{z} = e_3^+ \le x_i$.*

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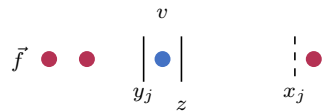
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Proof

- **Base.** $Z(0) = \varnothing$.
- **Step.** *Every older inserted switch keeps its witness: the Y-switches stay fixed.*
- *Y5 gives $y_i = e_3^- < x_i$. The next possible switch position is e_3^+, so $\hat{z} = e_3^+ \le x_i$.*
- *Take $v = e_3$ and $j = i$.*

Z4: two switches bracket a vertex



An inserted z occurs only immediately right of v:
$y_j = v^-, \quad z = v^+ \le x_j.$

Z1–Z4: termination gives a proper 2-colouring

Z2. $M(c_{Z,t}) \subsetneq M(c_{Z,t-1})$ for $t \geq 1$.
Z3. Every monochromatic edge in $c_{Z,t}$ is a $(2,3)$-edge, with $z < \vec{f}$ for all $z \in Z(t)$.

Claim
c_Z is a proper 2-colouring.

Proof
- *Z2: the finite set of monochromatic edges strictly shrinks at every step.*

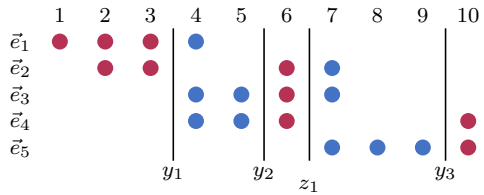
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- *Z2: the finite set of monochromatic edges strictly shrinks at every step.*
- *At termination no $(2,3)$-edge remains. By Z3, no monochromatic edge remains.*



Theorem: every $r \geq 4$ shift-chain is 2-colourable

X1. Every edge $\vec{f}$ has exactly one or two switches in c_X.
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Y6. Every monochromatic edge in c_Y is a $(2,3)$-edge.
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Theorem (Smith, 2026)
All r-shift-chains of uniformity at least $r_0 = 4$ are properly 2-colourable.

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- For $r > 4$, keep the first four vertices of each edge. These form a 4-uniform shift-chain; its proper colouring also colours every original edge properly.

Thank you!